

Broad-spectrum immune suppression encoded in self-amplifying RNA enables non-cytotoxic, non-immunostimulatory, externally controllable transgene expression

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eLife Assessment

In this manuscript, Lim and collaborators present a **useful** system for developing self-amplifying RNA that should not provoke a strong host inflammatory response. However, some of the claims are **incomplete**; additional experiments to investigate the effects on translation of the gene of interest and replication efficiency of the self-amplifying RNA could strengthen the manuscript.

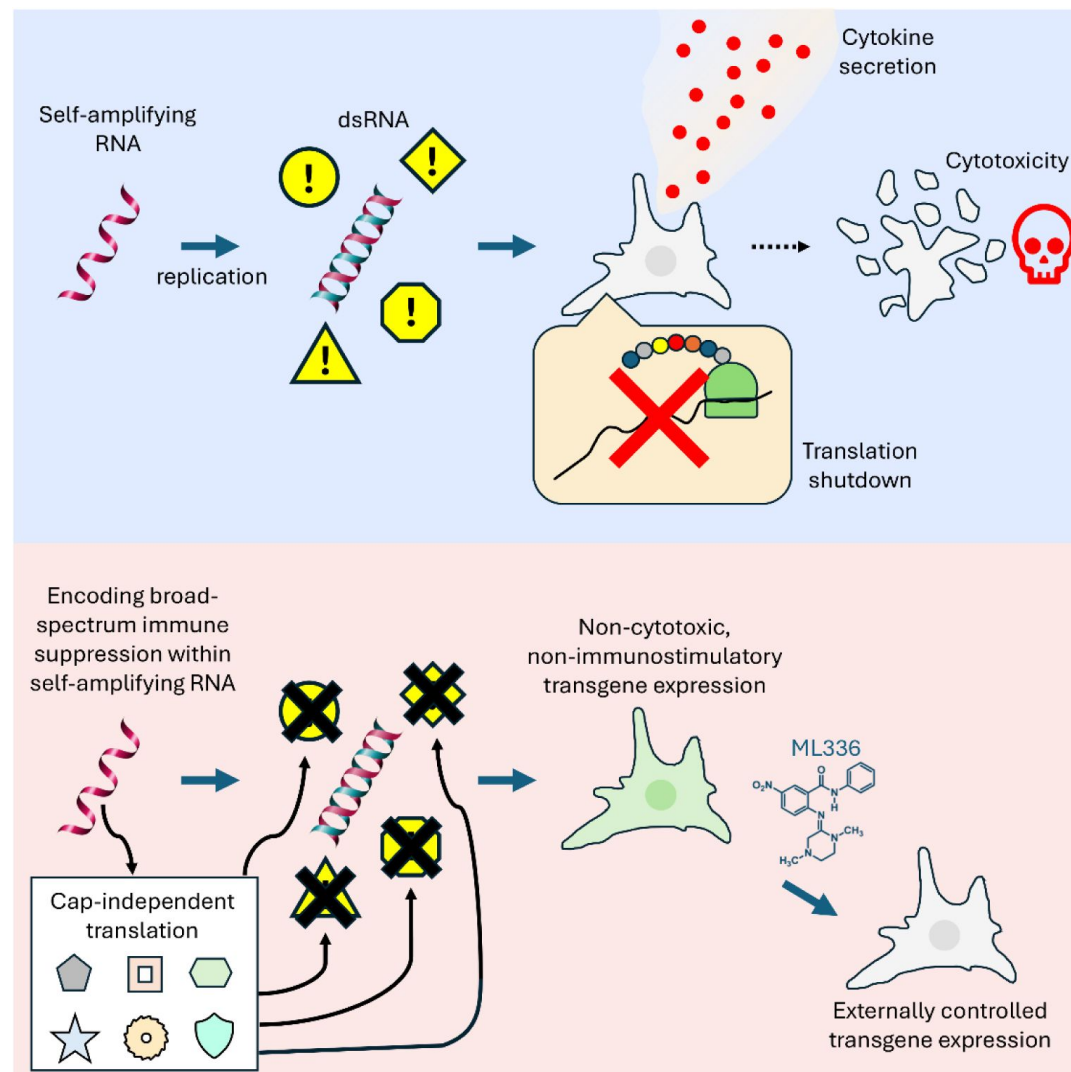
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Abstract

Self-amplifying RNA (saRNA) has the potential to provide durable, non-integrative transgene expression for transient gene therapy. However, its auto-replicative nature mimics viral infection, triggering innate immune responses that shutdown cap-dependent translation, degrade cellular mRNA, induce cell death, and release cytokines. In non-immunotherapy applications, this immune activation is undesirable as it limits transgene expression, depletes transfected cells, and induces inflammation, undermining therapeutic outcomes. Moreover, the use of exogenous immune suppressants to mitigate these effects often increases treatment complexity and the risk of unintended systemic side effects. To address these challenges, we developed a strategy to encode broad-spectrum innate immune suppression directly within saRNA. This approach leverages cap-independent translation to bypass saRNA-triggered translation shutdown, enabling the expression of multiple inhibitors targeting diverse double-stranded RNA-sensing and inflammatory signaling pathways. In mouse primary fibroblast-like synoviocytes—a cell type relevant to inflammatory joint diseases—this strategy eliminates the need for external immune inhibitors, reduces cytotoxicity and antiviral cytokine secretion, and enables sustained transgene expression that can be controlled with a small-molecule antiviral. These findings support the development of saRNA therapeutics that offer durable, non-integrative, externally controllable transgene

expression without persistent immune activation or reliance on exogenous immune suppressants.

Graphical abstract



Introduction

Self-amplifying RNA (saRNA) is a promising platform for transient gene therapy due to its ability to encode large therapeutic payloads and achieve sustained, non-integrative protein expression.¹ Upon cellular internalization, saRNA utilizes host machinery to synthesize an RNA-dependent RNA polymerase (RdRp), which drives replication of the saRNA genome.² This process involves the production of a negative-strand RNA intermediate that serves as a template for both genomic and subgenomic RNA synthesis.³ Subgenomic RNA, encoding the gene of interest, is transcribed in excess of the genomic RNA, leading to robust and sustained protein expression.⁴

However, the self-replication of saRNA also generates double-stranded RNA (dsRNA) intermediates, formed both during the synthesis of negative-strand RNA and during the transcription of positive-strand RNA from the negative-strand template.³ These dsRNA

intermediates are potent activators of cytosolic pattern recognition receptors, triggering innate immune responses.^{5,6} This recognition results in shutdown of cap-dependent translation,^{7–10} degradation of cellular mRNA,^{6,11} induction of programmed cell death,^{6,10,12–16} and release of cytokines and chemokines characteristic of viral infection.^{9,17,18} In non-immunotherapy applications such as transient gene therapy, these responses are particularly problematic, as they limit transgene expression, deplete transfected cells, and induce inflammation.

Strategies to mitigate saRNA-mediated innate immune responses have included the incorporation of modified nucleotides,¹⁹ removal of dsRNA contaminants,^{9,20} and co-delivery of non-replicating mRNA encoding viral innate immune inhibiting proteins.^{21–23} While these approaches can reduce initial innate immune responses to saRNA transfection, they fall short in addressing the continuous immune activation triggered by dsRNA intermediates that arise during saRNA replication.² Sequence evolution of the RdRp replicon shows some promise in reducing these responses, but even engineered replicons still induce significant innate immune activation.^{6,24} Encoding viral immune inhibitors within the saRNA construct using 2A self-cleaving peptides can improve protein expression *in vitro* but has limited impact on cytokine responses, despite reductions in NF-κB and IRF3 activation.¹⁸ Although these strategies temper saRNA-induced innate immune responses to levels appropriate for immunotherapy applications—where some immune activation is beneficial²⁵—they fall short of the stringent requirements for non-immunotherapy contexts, where therapeutic outcomes can be compromised by immune activation. In such cases, exogenous immunosuppressants can mitigate saRNA-induced immune responses and enable effective transgene expression,^{9,21,26,27} but this approach complicates treatment regimens and increases the risk of unintended side effects from systemic immune suppression.

To address these challenges, we developed a fully saRNA-based approach that mitigates innate immune responses triggered by saRNA replication, incorporating several key innovations. First, innate immune inhibitory proteins are expressed via cap-independent translation, bypassing the cap-dependent translation shutdown commonly triggered by saRNA. Cap-dependent translation requires the presence of a 5' cap structure on mRNA, which recruits translation initiation factors and ribosomes.²⁸ By contrast, cap-independent translation uses internal ribosome entry sites (IRES)—to recruit ribosomes directly to the mRNA, enabling translation even when cap-dependent pathways are inhibited.²⁹ Second, by simultaneously targeting multiple dsRNA-sensing and inflammatory signaling pathways, our method provides a more comprehensive suppression of innate immune responses than single target approaches. Third, encoding innate immune inhibitors directly within the saRNA ensures continuous protection against persistent innate immune activation caused by dsRNA intermediates during saRNA replication, eliminating the need for exogenous immunosuppressants.

We evaluated this approach in mouse primary fibroblast-like synoviocytes (FLS), a key cell type in the pathology of inflammatory joint diseases.^{30,31} To facilitate this evaluation, we developed a microplate assay for straightforward, longitudinal monitoring of saRNA translation control and cytotoxicity in live cells. Our results demonstrate that this strategy effectively reduces saRNA-induced cytotoxicity and cytokine production while enabling sustained cap-dependent transgene expression, which can be reversed with a small-molecule antiviral. These results pave the way for saRNA-based therapies that offer durable, externally controllable transgene expression with minimal innate immune activation, achieving this without reliance on exogenous immunosuppressants.

Results

Development of a microplate assay for longitudinal monitoring of translational control and cell number

Since saRNA can induce cap-dependent translation shutdown and cell death, we developed a microplate assay to simultaneously monitor cap-independent translation, cap-dependent translation, and cell number over time. In this assay, primary mouse FLS (Fig. S1) were plated in 6-well plates and transfected with saRNA constructs that used the Venezuelan equine encephalitis virus (VEEV) RdRp to self-replicate.

These constructs featured a dual-fluorescence design, with EGFP expressed through encephalomyocarditis virus (EMCV) IRES-mediated, cap-independent translation, and mScarlet3 expressed through subgenomic cap-dependent translation. This dual-reporter system enabled assessment of the effects of saRNA on translational control (**Fig. 1a** [↗](#)). Spectral overlap between EGFP and mScarlet3 was corrected using linear unmixing (Figs. S2a-c).

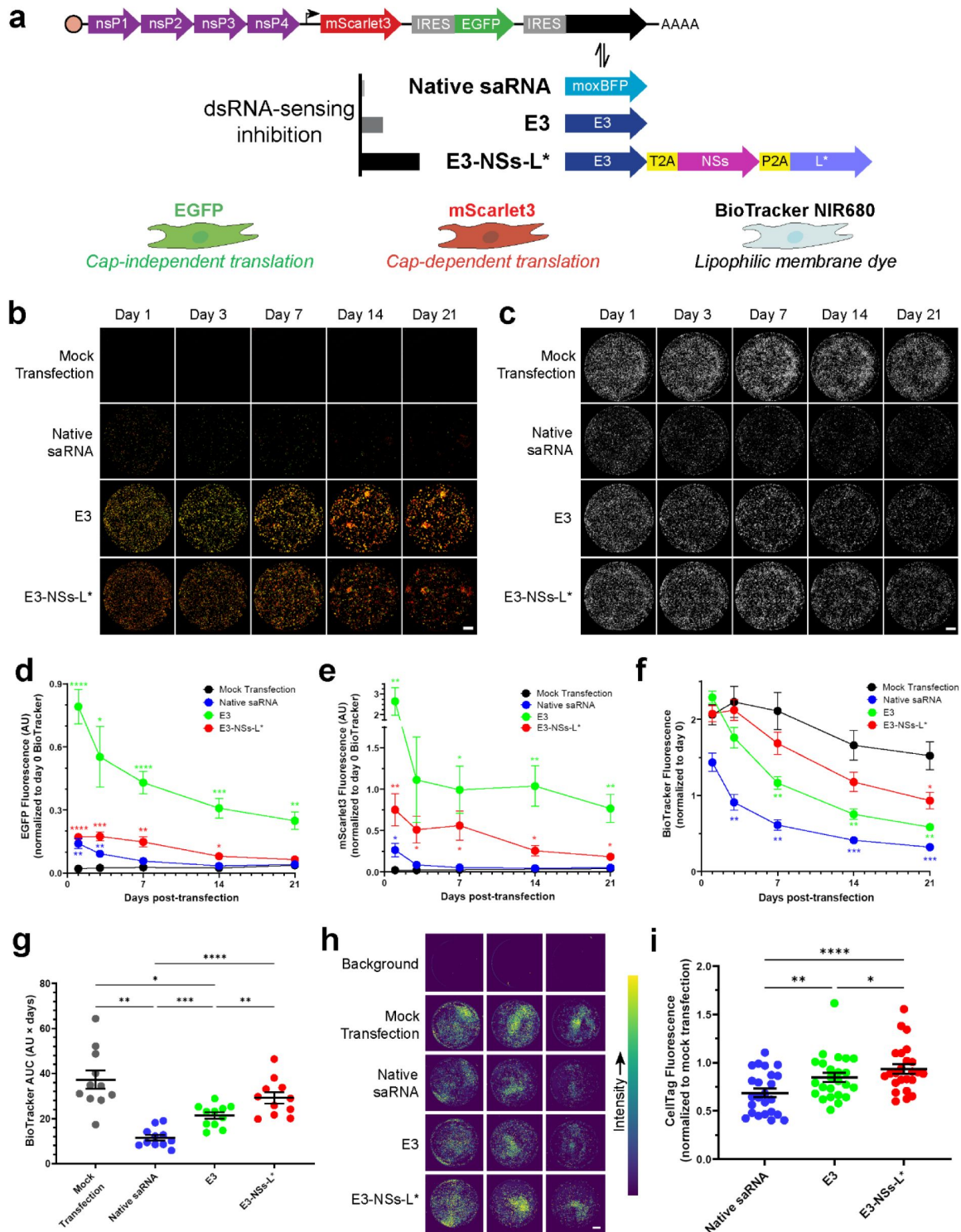


Fig. 1.

Differential effects of moderate and strong dsRNA-sensing pathway inhibition on saRNA transgene expression and cell loss.

a, Schematic of the native saRNA, E3, and E3-NSs-L* constructs designed to inhibit dsRNA-sensing pathways and report saRNA transgene expression. The native saRNA construct lacks dsRNA-sensing inhibitors. The E3 construct expresses vaccinia virus E3, a pleiotropic inhibitor of dsRNA sensing, expected to provide moderate inhibition. The E3-NSs-L* construct expresses vaccinia virus E3, and additionally includes Toscana virus NSs and Theiler's virus L*, which target the PKR and OAS/RNase L pathways, expected to provide strong inhibition. EGFP is expressed via an IRES to report cap-independent translation, while a subgenomic promoter (depicted with an angled arrow) enables transcription of an RNA transcript that expresses mScarlet3 via cap-dependent translation. saRNA constructs were transfected into primary mouse FLS, which were labeled with BioTracker to monitor cell number.

b, Representative images of EGFP (green) and mScarlet3 (red) expression in FLS transfected with native saRNA, E3, or E3-NSs-L* over 3 weeks. Scale bar = 5 mm.

c, Representative images of FLS transfected with the same constructs, showing BioTracker intensity over time. Scale bar = 5 mm.

d, Quantification of EGFP fluorescence over time (n = 11 biological replicates). The E3 construct provided the greatest EGFP expression, while the E3-NSs-L* construct showed intermediate levels. Statistical significance of treatment effects at each time point compared to mock transfection was determined by two-way RM ANOVA with Greenhouse–Geisser correction and Dunnett's multiple comparisons test.

e, Quantification of mScarlet3 fluorescence over time (n = 11 biological replicates). The E3 construct provided the greatest mScarlet3 expression, while the E3-NSs-L* showed intermediate levels. Statistical significance of treatment effects at each time point compared to mock transfection was determined by two-way RM ANOVA with Greenhouse–Geisser correction and Dunnett's multiple comparisons test.

f, Quantification of BioTracker fluorescence over time (n = 11 biological replicates). The native saRNA and E3 constructs reduced BioTracker fluorescence, indicating cell loss. Statistical significance of treatment effects at each time point compared to mock transfection was determined by two-way RM ANOVA with Greenhouse–Geisser correction and Dunnett's multiple comparisons test.

g, AUC analysis of BioTracker fluorescence data shown in (f), summarizing cumulative effects over the time course (n = 11 biological replicates). Increasing dsRNA-sensing pathway inhibition prevents saRNA-induced reductions in integrated BioTracker signal. Statistical analysis was performed using one-way RM ANOVA with Greenhouse–Geisser correction and Tukey's multiple comparisons test to compare all groups. Mock transfection data is also presented in Fig. S5a.

h, Representative images of FLS stained with the cell number normalization dye, CellTag 700. Columns represent cells plated from the same biological replicate, while rows show different treatments. Scale bar = 2.5 mm.

i, Quantification of mock transfection normalized CellTag signal (n = 24 biological replicates). Increasing dsRNA-sensing pathway inhibition mitigates saRNA-induced reductions in CellTag signal. Statistical analysis was performed using one-way RM ANOVA with Greenhouse–Geisser correction and Holm–Šidák's multiple comparisons test to compare all groups. Assays were performed 2 days post-transfection. Data in this panel were pooled from in-cell western assays, incorporating data presented in [Figs. 3](#) and [6](#) as well as additional data not shown elsewhere.

For all statistical reporting, *P < 0.05, **P < 0.01, ***P < 0.001 and ****P < 0.0001. Data are presented as mean ± SEM.

For panels (d-f): Data were normalized to starting cell number, indicated by BioTracker intensity on day 0, prior to transfection. The mock transfection control data is also presented in [Figs. 5c-e](#).

Acronyms: saRNA, self-amplifying RNA; dsRNA, double-stranded RNA; nsP, Venezuelan equine encephalitis virus non-structural protein; IRES, encephalomyocarditis virus internal ribosome entry site; moxBFP, monomeric oxidizing environment-optimized blue fluorescent protein; E3, Vaccinia virus E3 protein; NSs, Toscana virus non-structural NSs protein; L*, Theiler's murine encephalomyelitis virus L* protein; T2A, Thossea asigna virus 2A peptide; P2A, porcine teschovirus-1 2A peptide; PKR, protein kinase R; OAS, oligoadenylate synthase; AUC, area under the curve.

To monitor cell number, FLS were stained with BioTracker NIR680, a lipophilic carbocyanine membrane dye that enables long-term fluorescent labeling of cells.³² Carbocyanine dyes like BioTracker are weakly fluorescent in aqueous solution but exhibit high fluorescence when integrated into lipid bilayers.

Consequently, mock transfection with Lipofectamine MessengerMAX—a lipid transfection agent—enhanced BioTracker fluorescence compared to non-transfected cells, likely due to an interaction with the lipid components of Lipofectamine (Figs. S3a, S3b). Despite this interaction, BioTracker remains effective for indicating cell number, as treatment of mock transfected cells with the apoptosis inducer staurosporine³³ reduced BioTracker fluorescence (Figs. S3a, S3b). The BioTracker results were consistent with those obtained using the viability dye calcein AM (Figs. S3c, S3d), with the added advantages of eliminating the need for additional staining and wash steps and avoiding spectral overlap with red and green fluorescent proteins. This assay allows for seamless longitudinal monitoring of cap-independent translation, cap-dependent translation, and cell number using microplate readers or imagers.

Design of saRNA constructs for inhibiting multiple dsRNA sensing pathways

Cells detect cytosolic dsRNA through several sensing pathways, including RIG-I,³⁴ melanoma differentiation-associated protein 5 (MDA-5),³⁵ protein kinase R (PKR),³⁶ and the oligoadenylate synthase (OAS)/RNase L pathway.³⁷ To broadly inhibit these pathways, we designed an saRNA construct, referred to as 'E3', that utilizes EMCV IRES cap-independent translation to express the vaccinia virus E3 protein ([Fig. 1a](#)). E3 protein is a pleiotropic inhibitor that binds and sequesters dsRNA, effectively blocking multiple dsRNA sensing pathways.³⁸

Given the key roles of PKR in cap-dependent translation shutdown and OAS/RNase L in cellular mRNA degradation in response to dsRNA, we sought to further inhibit these pathways in a second construct. This construct includes Toscana virus NSs, a ubiquitin ligase that promotes PKR degradation,³⁹ and Theiler's virus L*, which inhibits RNase L.⁴⁰ To enable co-expression of these three proteins—E3, NSs, and L*—we incorporated 2A “self-cleaving” peptide sequences between the proteins, allowing the polyprotein to be cleaved into separate proteins during translation. As prior experiments showed that using multiple identical 2A peptides can reduce protein expression,⁴¹ we selected non-identical 2A peptides for this construct. This construct, named 'E3-NSs-L*', is expected to provide a more comprehensive inhibition of dsRNA-sensing pathways than the E3 construct.

Finally, as a control, we designed a 'native saRNA' construct that expresses a blue fluorescent protein (moxBFP) as a visual marker of IRES functionality in place of inhibitors of dsRNA-sensing pathways. Importantly, moxBFP expression did not overlap spectrally with the EGFP or mScarlet3 channels (Figs. S2a-c).

Moderate dsRNA-sensing pathway inhibition enables high transgene expression at the cost of cell loss, while strong inhibition preserves cell number at lower expression levels

Following transfection of saRNA constructs, we monitored BioTracker, EGFP, and mScarlet3 fluorescence over 3 weeks (**Figs. 1b, 1c**). Inhibiting dsRNA-sensing pathways with viral proteins significantly enhanced saRNA transgene expression (**Figs. 1d, 1e**). Interestingly, the E3 construct produced the highest levels of EGFP and mScarlet3 expression, surpassing both native saRNA and the E3-NSs-L* construct. Native saRNA yielded low levels of transgene expression, while E3-NSs-L* showed intermediate expression.

Achieving durable transgene expression depends equally on preserving cell viability and maximizing transgene expression, as constructs that induce cell death undermine the goal of sustained transgene expression. Transfection of FLS with native saRNA caused both immediate and long-term reductions in cell number, as indicated by BioTracker intensity (**Fig. 1f**). While the E3 construct initially maintained cell number, BioTracker signals gradually diminished over time. In contrast, the E3-NSs-L* construct provided sustained protection against the decline in cell number. To quantify the cumulative effects of different saRNA constructs on cell number over the experimental time course, we performed area under the curve (AUC) analysis, demonstrating that E3-NSs-L* significantly protected against a decline in cell number compared to the E3 construct (**Fig. 1g**). These findings were further supported by experiments using CellTag, a cell number normalization stain (**Fig. 1h**), which revealed that increasing dsRNA-sensing pathway inhibition protects against cell loss (**Fig. 1i**). Together, these results suggest that while the E3 construct effectively enhances saRNA transgene expression, it is associated with greater cell loss, underscoring the advantage of the E3-NSs-L* construct for applications requiring sustained, non-cytotoxic gene expression.

Inhibiting dsRNA sensing pathways reduces saRNA-induced cytotoxicity and improves cell viability

saRNA is known to induce programmed cell death,^{6,10,12–16} which often leads to cell detachment.⁴² Thus reductions in BioTracker signal following saRNA treatment are likely indicative of this process. However, since BioTracker is a lipophilic membrane dye, the reduction in signal may not necessarily correlate with cell viability or cytotoxicity. In addition to long-term cell tracking, lipophilic membrane dyes like BioTracker are often used to stain extracellular vesicles, which bud off from the plasma membrane.⁴³ During cellular stress, increased production of extracellular vesicles⁴⁴ could deplete the membrane-associated BioTracker dye from cells, resulting in a lower detectable signal. Therefore, the observed reduction in BioTracker signal could reflect enhanced extracellular vesicle production rather than cell death.

To provide more definitive evidence that inhibiting dsRNA-sensing pathways protects against saRNA-induced cytotoxicity, we conducted a longitudinal assay with annexin V staining over 6 days (**Fig. 2a**), followed by calcein AM staining on day 7 (**Fig. 2b**). Annexin V, a membrane-impermeable protein, binds to phosphatidylserine, which translocates to the extracellular side of the cell membrane during early apoptosis, enabling the detection of apoptotic cells with intact membranes.⁴⁵ If the membrane is compromised, annexin V can enter cells and stain intracellular phosphatidylserine, indicating late apoptosis or necrosis.⁴⁶ Calcein AM, a viability dye, accumulates in metabolically active cells with intact membranes, serving both as a marker of live cell number and an indicator of membrane integrity.⁴⁷

The apoptosis inducing agent staurosporine caused a transient increase in annexin V staining that decreased after the first day (**Fig. 2c**), along with a marked reduction in calcein AM fluorescence compared to mock transfection (**Fig. 2d**), observations that are consistent with extracellular

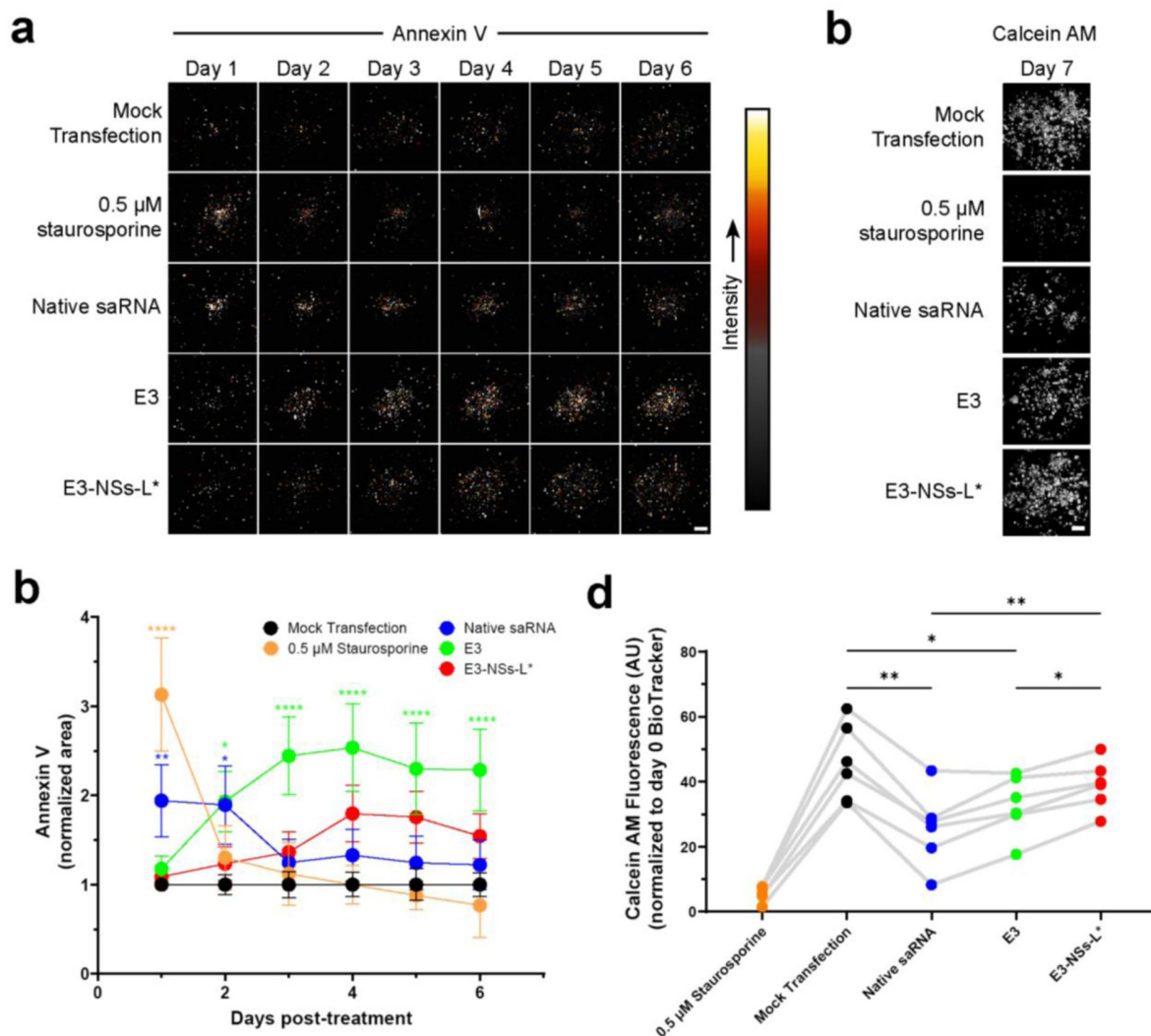


Fig. 2.

saRNA induces increased phosphatidylserine staining and reduced viability, which is prevented by E3-NSs-L*.

a, Representative cropped images of Annexin V-CF800 staining, indicating phosphatidylserine exposure or loss of membrane integrity, performed daily over 6 days using a microplate imager. Scale bar = 1.5 mm.

b, Representative cropped images of calcein AM staining, indicating viability, on day 7 post-treatment. Scale bar = 1.5 mm.

c, Annexin V staining, quantified as the area of positive pixels determined using Li thresholding ($n = 6$ biological replicates). Staurosporine, native saRNA, and the E3 construct significantly increased annexin V staining, while E3-NSs-L* did not. Data are normalized to the average of the mock transfection group. Statistical significance of treatment effects at each time point compared to mock transfection was determined using two-way RM ANOVA with Bonferroni's multiple comparisons test. Data are presented as mean $\pm$ SEM.

d, Calcein AM intensity measured on day 7 post-treatment ($n = 6$ biological replicates). Native saRNA, and E3 significantly reduced cell viability compared to mock transfection, while E3-NSs-L* did not. Cell viability in the E3-NSs-L* group was significantly higher than the E3 group. Connecting lines indicate responses from the same biological replicate. All groups differed significantly from staurosporine; these comparisons are omitted from the figure for clarity due to the large number of statistical comparisons. Data are normalized to cell number on day 0 as determined by BioTracker staining before transfection. Statistical significance was determined by one-way RM ANOVA with Greenhouse-Geisser correction and Tukey's multiple comparisons test comparing all groups.

saRNA constructs used are shown in [Fig. 1a](#). For all statistical reporting, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ and **** $P < 0.0001$.

phosphatidylserine translocation and/or increased membrane permeability. Similarly, native saRNA induced a temporary increase in annexin V staining during the first two days post-transfection (**Fig. 2c**) accompanied by reduced calcein AM staining (**Fig. 2d**). Co-expression of E3 prevented the initial increase in annexin V staining, but elevated it from day 2 onward, and also resulted in significantly reduced calcein AM intensity (**Figs. 2c, 2d**). In contrast, E3-NSs-L* had no significant effect on annexin V or calcein AM staining compared to mock transfection, supporting the idea that broader inhibition of dsRNA-sensing pathways, achieved by combining E3, NSs and L*, is more effective than E3 alone in preventing saRNA-induced cell death and preserving cell viability.

E3 and E3-NSs-L* prevent saRNA-induced eIF2 α phosphorylation and differentially regulate eIF2 α levels

After characterizing how moderate and strong inhibition of dsRNA-sensing pathways affects saRNA-induced cytotoxicity, we next investigated molecular mechanisms of translational control to understand how this inhibition influences transgene expression. Eukaryotic initiation factor 2 alpha (eIF2 α) is a central regulator of translation initiation; its phosphorylation serves as a cellular stress response to limit cellular cap-dependent protein synthesis, freeing ribosomes for cap-independent translation initiation instead.⁴⁸ Using in-cell western assays, we found that native saRNA transfection increased eIF2 α phosphorylation, but both E3 and E3-NSs-L* effectively blocked this phosphorylation (**Fig. 3a**), consistent with previous reports of vaccinia virus E3's inhibitory effect on eIF2 α phosphorylation.²² Interestingly, E3-NSs-L* also increased total eIF2 α levels, an effect not observed with the E3 construct (**Fig. 3b**).

E3-NSs-L* counteracts saRNA-induced PKR upregulation

Given that PKR phosphorylates eIF2 α upon activation by dsRNA,⁷ we next assessed its levels following saRNA transfection. Consistent with previous reports,⁶ native saRNA transfection increased PKR levels (**Fig. 3c**). The E3 construct similarly increased PKR levels; however, co-expression of E3-NSs-L* mitigated this increase, suggesting that E3-NSs-L* effectively counteracts saRNA-induced PKR upregulation. This is in line with the known role of the Toscana virus NSs protein as a ubiquitin ligase that targets PKR for degradation.³⁹

saRNA reduces eIF4E phosphorylation without altering total eIF4E levels

We next examined the impact of saRNA transfection on eIF4E, a key regulator of translation initiation.⁴⁹ All constructs reduced eIF4E phosphorylation levels (**Fig 3d**), while total eIF4E levels remained unchanged (**Fig. 3e**).

E3 reduces rRNA integrity, while E3-NSs-L* protects against this reduction

eIF2 α -and eIF4E-mediated translational control do not account for the greater transgene expression observed with the E3 construct compared to the E3-NSs-L* construct. This is evidenced by the lack of significant differences in eIF2 α and eIF4E phosphorylation status or levels between the two constructs (**Figs. 3a-b**, 3c-e), suggesting that additional mechanisms beyond eIF2 α -and eIF4E-mediated control contribute to the enhanced transgene expression seen with the E3 construct.

When OAS binds dsRNA, it synthesizes 2'-5'-oligoadenylate, which activates RNase L to cleave single-stranded RNA, leading to cellular mRNA degradation⁵⁰ and inhibition of mRNA nuclear export,⁵¹ thereby massively reducing host protein synthesis.⁵² Many RNA viruses, including dengue and Zika, evade RNase L by localizing their transcripts within viral replication organelles,

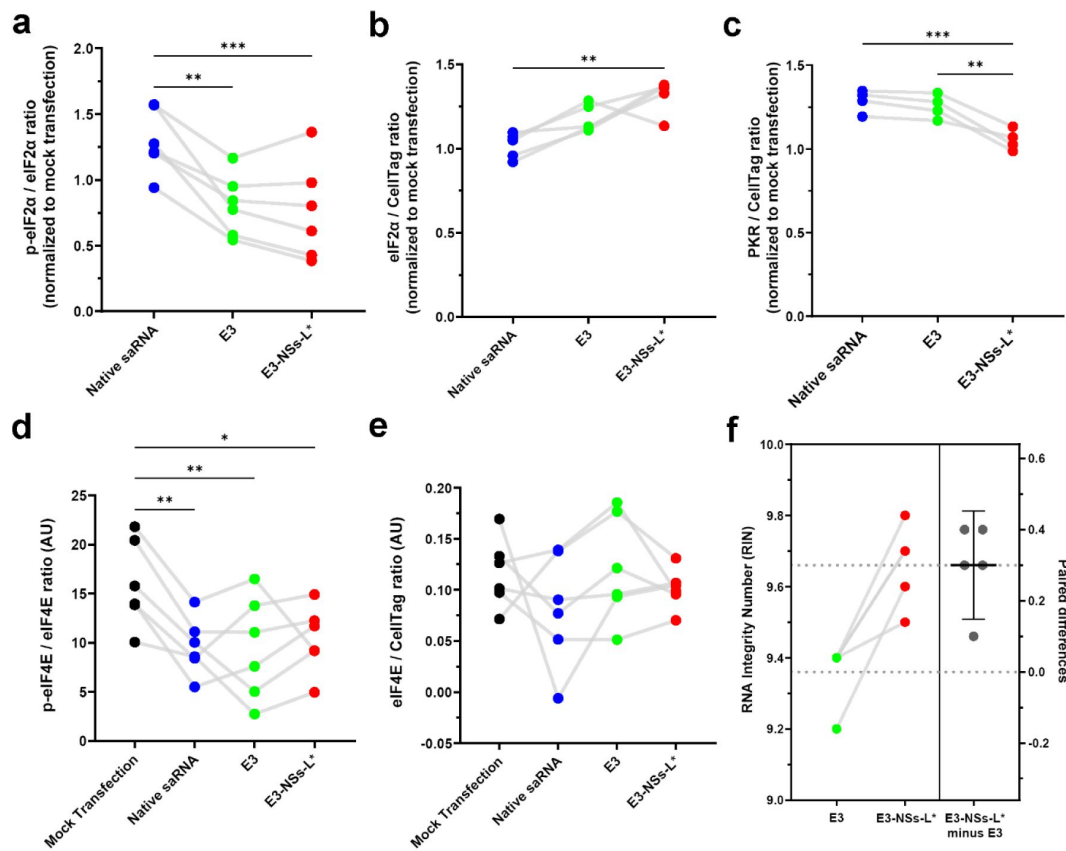


Fig. 3.

Translational control alterations induced by saRNA are modulated by dsRNA-sensing pathway inhibitors with differing effects on RNA integrity.

a, Phosphorylated eIF2 α levels examined day 2 post-transfection by in-cell western assay (n = 6 biological replicates). Both E3 and E3-NSs-L* constructs significantly reduced eIF2 α phosphorylation. Data are presented as fold-change relative to mock transfected cells. Statistical significance was determined by one-way RM ANOVA with Tukey's multiple comparisons test.

b, eIF2 α levels examined day 2 post-transfection by in-cell western assay (n = 5 biological replicates). E3-NSs-L* significantly increased total eIF2 α levels compared to native saRNA transfection. Data are presented as fold-change relative to mock transfected cells. Statistical significance was determined by one-way RM ANOVA with Tukey's multiple comparisons test.

c, PKR levels examined day 2 post-transfection by in-cell western assay (n = 4 biological replicates). E3-NSs-L* significantly reduced PKR levels compared to both native saRNA and E3 transfection. Data are presented as fold-change relative to mock transfected cells. Statistical significance was determined by one-way RM ANOVA with Tukey's multiple comparisons test.

d, Phosphorylated eIF4E levels examined day 2 post-transfection by in-cell western assay (n = 6 biological replicates). All saRNA constructs tested significantly reduced eIF4E phosphorylation levels compared to mock transfection. Statistical significance was determined by one-way RM ANOVA with Tukey's multiple comparisons test.

e, eIF4E levels examined day 2 post-transfection by in-cell western assay (n = 6 biological replicates). One-way RM ANOVA revealed no significant differences between groups (F(3,15) = 1.207, P = 0.3410). **f**, rRNA integrity of FLS transfected with E3-NSs-L* is significantly higher than that of E3-transfected FLS (n = 5 biological replicates). rRNA integrity was assessed using the RNA Integrity Number (RIN) algorithm, which ranges from 1 to 10, with 10 indicating fully intact rRNA. Total RNA was extracted from FLS 1 day post-transfection. Data are shown as a Gardner-Altman comparison plot, with the right panel illustrating the mean effect size $\pm$ 95% CI. Statistical significance was assessed using a paired t-test (P = 0.0054).

saRNA constructs used are shown in [Fig. 1a](#). Connecting lines indicate responses from the same biological replicate. Statistical significance is indicated as follows: *P < 0.05, **P < 0.01, ***P < 0.001. The mock transfection control data presented in panels (a), (b), (d) and (f) are also used in [Fig. 6](#).

protecting them from degradation while allowing continued protein production.^{51,53} Similarly, the VEEV-derived saRNA used in this study replicates within these organelles,⁵⁴ likely protecting its transcripts from RNase L activity. Consequently, RNase L activation by the E3 construct may selectively deplete cytoplasmic host mRNA while sparing saRNA. This depletion reduces competition for ribosomes, thereby enhancing the translation of saRNA transcripts. In contrast, the E3-NSs-L* construct encodes Theiler's virus L*, an RNase L inhibitor,⁴⁰ thereby preventing host mRNA degradation and forcing saRNA transcripts to compete directly with cellular mRNA for ribosomal access.

To determine whether RNase L activation contributes to the greater transgene expression observed with the E3 construct compared to E3-NSs-L*, we assessed RNase L activity by measuring ribosomal RNA (rRNA) cleavage, an indicator of RNase L activity.⁵⁵ RNA extracted from FLS transfected with the E3 and E3-NSs-L* constructs was evaluated with the RNA integrity number (RIN) algorithm, an automated Bayesian learning method for assessing rRNA integrity.⁵⁶ Cells transfected with E3 exhibited significantly lower rRNA integrity than those transfected with E3-NSs-L* (**Fig. 3f**). As prior studies have shown that RNase L activation primarily depletes host mRNA levels before extensive rRNA degradation,^{50,52,57} our findings support the hypothesis that RNase L activation and subsequent reductions in cellular mRNA occur with the E3 construct but not with E3-NSs-L*.

Design of saRNA constructs for inhibiting multiple inflammatory signaling pathways

Having addressed key saRNA-induced toxicities—including cell death, translation shutdown, and mRNA degradation—with the E3-NSs-L* construct, we next sought to design new saRNA constructs aimed at mitigating cytokine production. In these new constructs, E3-NSs-L*-EGFP was expressed via a single IRES using non-identical 2A peptides (**Fig. 4a**). On a second IRES, varying amounts of cellular inhibitors of inflammatory signaling were expressed. The control construct, called 'moxBFP', lacks cellular inhibitors of inflammatory signaling and instead expresses a blue fluorescent protein.

Given that activation of nuclear factor- κ B (NF- κ B) is a hallmark of viral infection and a master regulator of cytokine and chemokine induction,^{58,59} we prioritized its inhibition using super repressor inhibitor of κ Ba (srI κ Ba). This dominant active form of I κ Ba cannot be phosphorylated and thus resists ubiquitination and degradation, forming a stable cytoplasmic pool of I κ Ba that prevents NF- κ B nuclear translocation and downstream signaling.^{60,61} We generated one construct expressing srI κ Ba alone (named 'srI κ Ba') and another expressing srI κ Ba in combination with Smad7 and suppressor of cytokine signaling 1 (SOCS1) using non-identical 2A peptides (named 'srI κ Ba-Smad7-SOCS1'). Smad7 negatively regulates both transforming growth factor- β (TGF- β) and NF- κ B signaling pathways,⁶² while SOCS1 inhibits type I, II, and III interferon (IFN) pathways as well as NF- κ B signaling.^{63–65}

Inhibiting dsRNA-sensing pathways and inflammatory signaling suppresses saRNA-induced cytokine secretion

Using a bead-based immunoassay on FLS culture supernatants, we observed that native saRNA transfection induced cytokine responses associated with viral infection (**Figs. 4b**, **S4a-m**). Cytokine secretion was still apparent when dsRNA-sensing pathways were inhibited, although a reduction in tumor necrosis factor (TNF)- α , IFN- α , and IFN- β levels were observed with E3 expression. Similarly, C-X-C motif chemokine ligand 10 (CXCL10) levels were reduced with E3-NSs-L* expression. On the other hand, srI κ Ba co-expression broadly suppressed cytokine secretion, reducing all measured cytokines to levels comparable to mock-transfected FLS. The srI κ Ba-Smad7-SOCS1 construct did not further reduce cytokine levels beyond the suppression achieved by srI κ Ba alone.

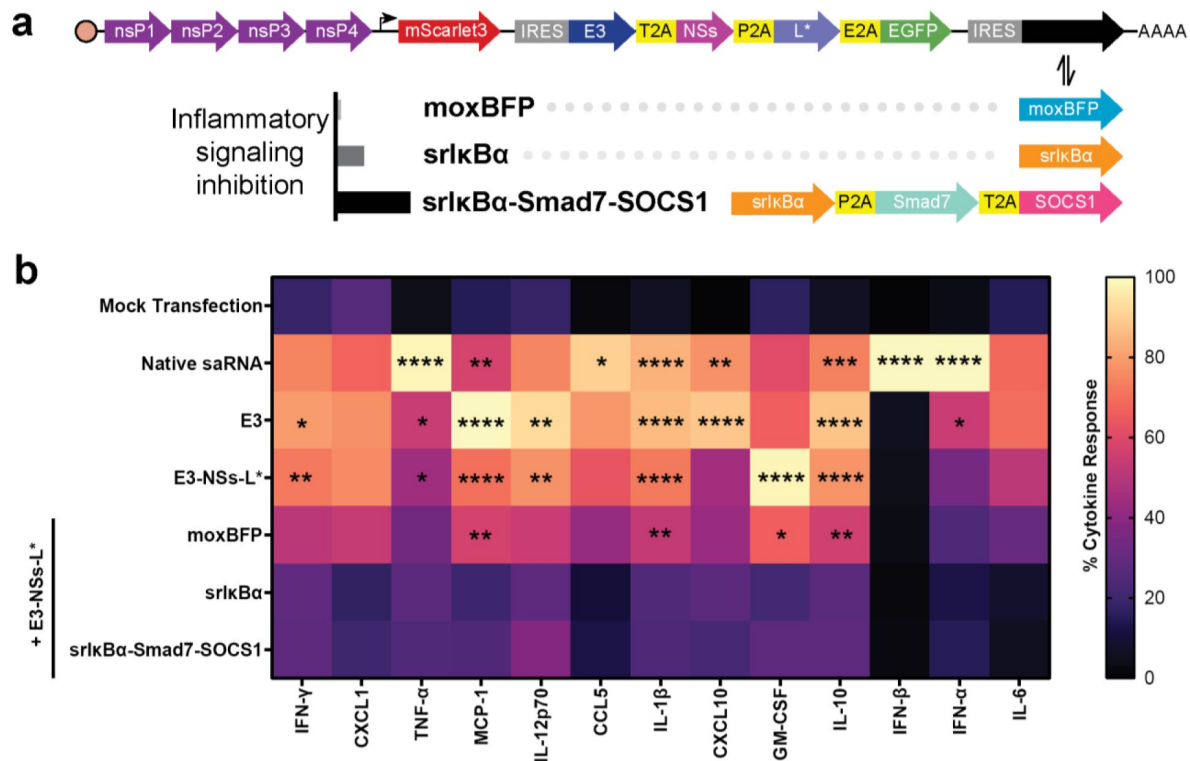


Fig. 4.

Inhibiting inflammatory signaling reduces saRNA-induced anti-viral cytokine secretion

a, Schematic of the moxBFP, srIkBa, and srIkBa-Smad7-SOCS1 saRNA constructs designed for inhibition of inflammatory signaling. These constructs include dsRNA-sensing pathway inhibitors (vaccinia virus E3, Toscana virus NSs, and Theiler's virus L*). The moxBFP construct, serving as a control, lacks inflammatory signaling inhibitors. The srIkBa construct co-expresses srIkBa, which blocks the NF-κB inflammatory signaling pathway and represents moderate inflammatory signaling inhibition. The srIkBa-Smad7-SOCS1 construct co-expresses srIkBa, Smad7 and SOCS1 to additionally inhibit TGF-β and IFN pathways, representing strong inflammatory signaling inhibition. The angled arrow denotes the subgenomic promoter.

b, FLS were transfected with saRNA constructs, and anti-viral cytokines were quantified by bead-based immunoassay on day 2 post-transfection (n = 6 biological replicates). dsRNA-sensing pathway inhibition and inflammatory signaling inhibition significantly reduced cytokine secretion induced by saRNA. Cytokine levels were scaled within each biological replicate, with the highest value set to 100%, and the mean value displayed in the heatmap. Detailed graphs of individual cytokines are shown in Supplementary Fig. S4. For statistical analysis, cytokine levels were normalized to pre-transfection cell number (measured using BioTracker). Statistical significance for each cytokine was assessed using one-way RM ANOVA followed by Bonferroni's multiple comparisons test against the mock transfection control. A Bonferroni correction was applied to P values to account for multiple independent hypothesis testing. *P < 0.05, **P < 0.01, ***P < 0.001 and ****P < 0.0001.

Acronyms: nsP, Venezuelan equine encephalitis virus non-structural protein; IRES, encephalomyocarditis virus internal ribosome entry site; moxBFP, monomeric oxidizing environment-optimized blue fluorescent protein; E3, Vaccinia virus E3 protein; NSs, Toscana virus non-structural NSs protein; L*, Theiler's murine encephalomyelitis virus L* protein; T2A, Thossea asigna virus 2A peptide; P2A, porcine teschovirus-1 2A peptide; E2A, equine rhinitis A virus 2A peptide; srIkBa, super-repressor inhibitor of κB; smad7, mothers against decapentaplegic homolog 7; SOCS1, suppressor of cytokine signaling 1; IFN-γ, interferon-γ; CXCL1, C-X-C motif chemokine ligand 1; TNF-α, tumor necrosis factor-α; MCP-1, monocyte chemoattractant protein-1; IL-12p70, interleukin-12; CCL5, chemokine ligand 5; IL-1β, interleukin-1β; CXCL10, C-X-C motif chemokine ligand 10; GM-CSF, granulocyte-macrophage colony-stimulating factor; IL-10, interleukin-10; IFN-β, interferon-β; IFN-α, interferon-α; IL-6, interleukin-6.

srIkBa reduces cell number, whereas srIkBa-Smad7-SOCS1 preserves cell number and enhances transgene expression

We then examined the impact of inhibiting inflammatory signaling on cell number and transgene expression (Figs. 5a, 5b). Interestingly, the srIkBa construct caused both immediate and sustained reductions in cell number, as quantified by BioTracker signal, whereas the srIkBa-Smad7-SOCS1 construct maintained cell number comparable to mock transfection throughout the experiment (Fig. 5c). AUC analysis of the BioTracker data confirmed that srIkBa-Smad7-SOCS1 significantly mitigated the srIkBa-induced reduction in integrated BioTracker signal (Fig. S5a). Additional experiments using CellTag and calcein AM staining corroborated these findings, showing that srIkBa negatively impacted both cell number (Fig. S5b) and viability (Fig. S5c), effects that were prevented by co-expression of Smad7 and SOCS1.

All three constructs produced low levels of EGFP expression (Fig. 5d), which was expected given EGFP's position as the fourth protein in a quad-cistronic 2A element (Fig. 4a). Previous studies have demonstrated that protein expression decreases significantly for downstream genes in such configurations.⁴¹ Notably, while the moxBFP and srIkBa constructs generated intermediate mScarlet3 levels, the srIkBa-Smad7-SOCS1 construct produced 2-to 5-fold higher mScarlet3 fluorescence compared to the others (Fig. 5e).

Smad7 and SOCS1 co-expression prevents srIkBa-induced alterations in translational control

We next explored how srIkBa and srIkBa-Smad7-SOCS1 modulate translational regulation. Interestingly, srIkBa expression led to reductions in eIF2 α phosphorylation, a change that was prevented by co-expression of Smad7 and SOCS1 (Fig. 6a). While neither construct affected total eIF2 α levels or eIF4E phosphorylation compared to the moxBFP construct (Figs. 6b, 6c), srIkBa caused a significant reduction in total eIF4E levels (Fig. 6d). Given the critical role of eIF4E in cap-dependent translation, its depletion likely contributes to the poor transgene expression observed with srIkBa (Fig. 5e). These effects, combined with srIkBa-induced declines in cell number and viability (Figs. 5c, S5a-c), highlight the dual impact of srIkBa on translational machinery and cellular health. Importantly, Smad7 and SOCS1 co-expression preserved eIF4E levels, counteracting the disruption of cap-dependent translation caused by srIkBa.

Prolonged moxBFP transfection does not activate FLS, while srIkBa reduces basal activation

Fibroblast activation protein- α (FAP- α) is a serine protease expressed on the surface of FLS that contributes to extracellular matrix degradation and tissue remodeling.^{66,67} Its expression is minimal in normal adult FLS but increases significantly following inflammatory activation, such as in rheumatoid and osteoarthritis.^{68,69} Since viral infections can induce fibroblast activation,^{70,71} we hypothesized that prolonged saRNA transfection might similarly trigger FLS activation. To test this, we measured FAP- α levels using an in-cell western assay after 11 days of transfection. FLS transfected with moxBFP showed FAP- α levels comparable to those of mock-transfected cells, whereas srIkBa and srIkBa-Smad7-SOCS1 constructs resulted in significantly lower FAP- α levels compared to mock transfection (Figs. 7a, 7b).

These findings indicate that long-term saRNA transfection does not induce FLS activation, as measured by FAP- α levels, potentially alleviating concerns that saRNA might drive pathological fibroblast responses.

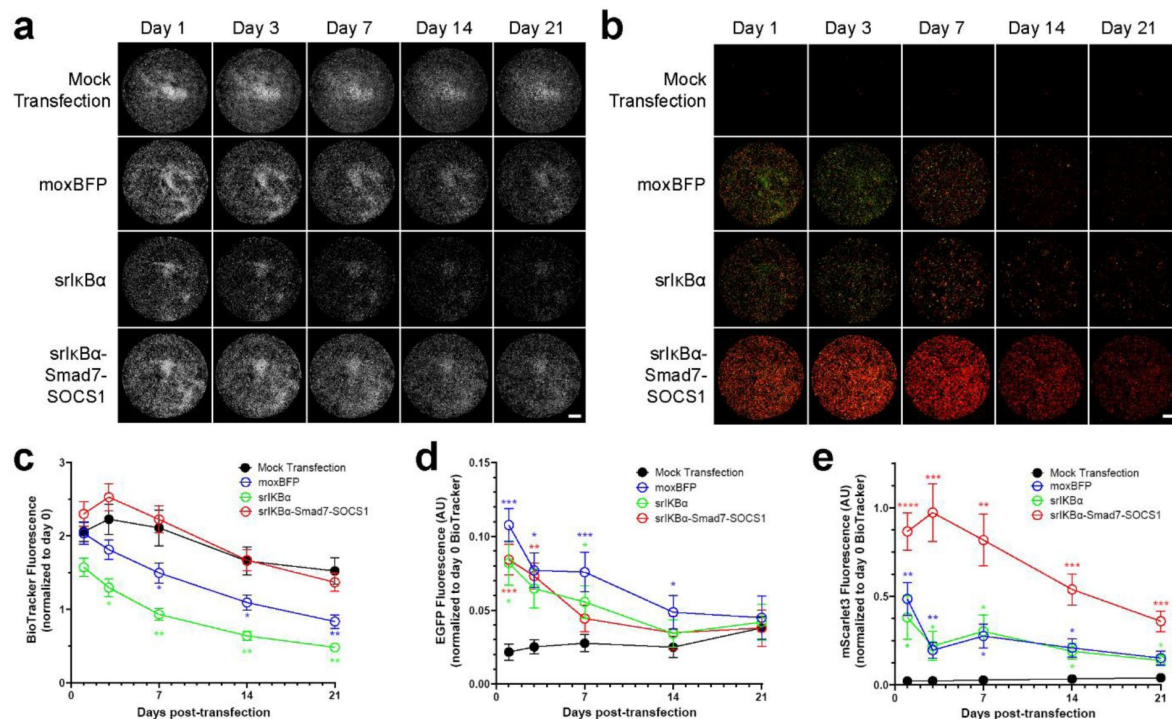


Fig. 5.

Differential effects of srIkBa and srIkBa-Smad7-SOCS1 on cell number and transgene expression.

a, Representative images of BioTracker staining over 3 weeks in FLS transfected with different saRNA constructs. Scale bar = 5 mm.

b, Representative images of EGFP (green) and mScarlet3 (red) expression over 3 weeks in FLS transfected with different saRNA constructs. Scale bar = 5 mm.

c, Quantification of BioTracker fluorescence intensity over time ($n = 11$ biological replicates). srIkBa induces reduction in cell number, which is prevented by srIkBa-Smad7-SOCS1. Statistical significance of treatment effects at each time point compared to mock transfection was determined by two-way RM ANOVA with Greenhouse-Geisser correction and Dunnett's multiple comparisons test.

d, Quantification of EGFP fluorescence intensity over time ($n = 11$ biological replicates). All constructs showed low levels of EGFP expression. Statistical significance of treatment effects at each time point compared to mock transfection was determined by two-way RM ANOVA with Greenhouse-Geisser correction and Dunnett's multiple comparisons test.

e, Quantification of mScarlet3 fluorescence intensity over time ($n = 11$ biological replicates). The srIkBa-Smad7-SOCS1 produced 2-5 times more mScarlet3 fluorescence than either moxBFP or srIkBa constructs. Statistical significance of treatment effects at each time point compared to mock transfection was determined by two-way RM ANOVA with Greenhouse-Geisser correction and Dunnett's multiple comparisons test.

saRNA constructs used are shown in [Fig. 4a](#). For all statistical reporting, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$ and $****P < 0.0001$. Data were normalized to cell number (BioTracker intensity) on day 0, prior to transfection. The mock transfection control data used in this figure is also presented in [Fig. 1](#). Data are presented as mean $\pm$ SEM.

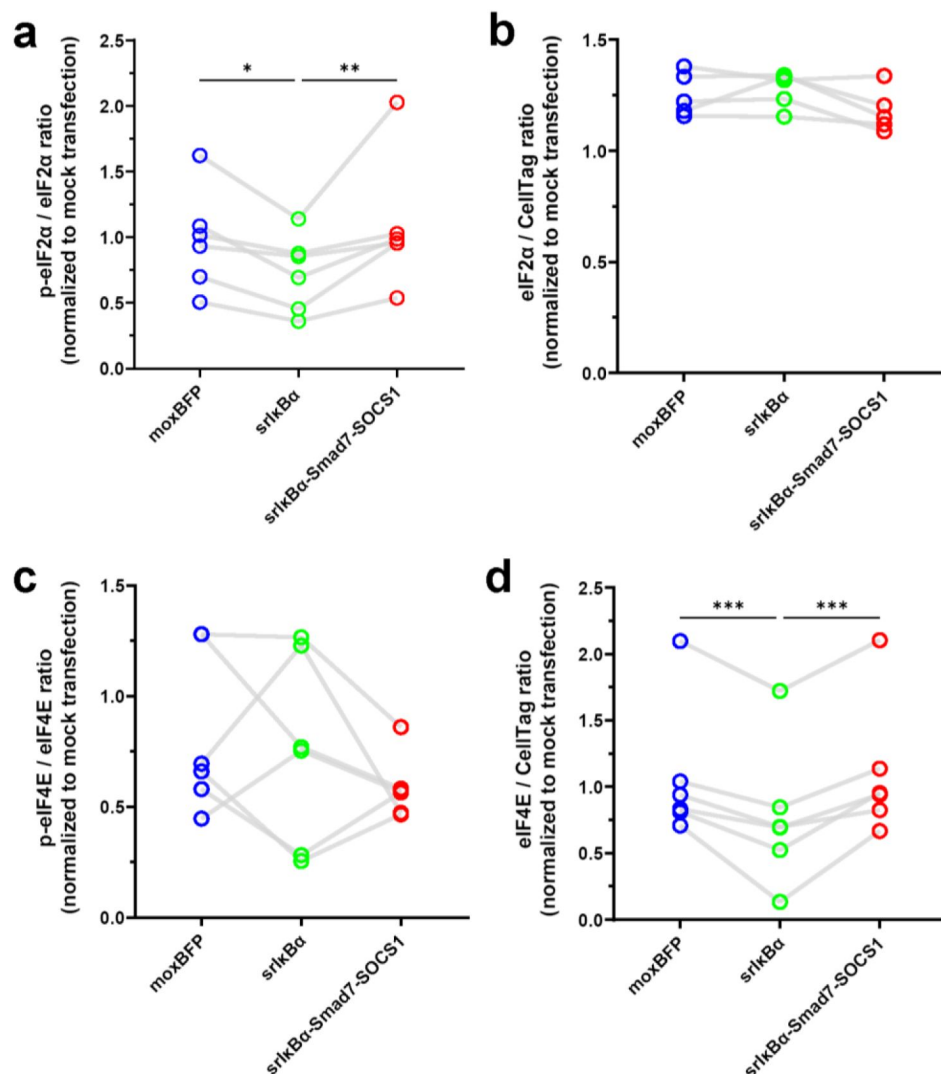


Fig. 6.

srIkBa reduces eIF2α phosphorylation and total eIF4E levels, effects reversed by co-expression of Smad7 and SOCS1 In-cell western assays were performed 2 days post-transfection. Data are presented as fold change relative to mock-transfected cells.

a, Phosphorylation of eIF2α is significantly reduced by srIkBa, and this reduction is reversed by co-expression of Smad7 and SOCS1 (n = 6 biological replicates). Statistical significance was determined by one-way RM ANOVA and Holm-Šidák's multiple comparisons test to compare all groups.

b, Total eIF2α levels are not significantly affected by srIkBa or srIkBa-Smad7-SOCS1 (n = 6 biological replicates). One-way RM ANOVA revealed no significant differences among groups. $F(2,8)=3.683$, $P=0.0735$.

c, Phosphorylation of eIF4E is not significantly affected by srIkBa or srIkBa-Smad7-SOCS1 (n = 6 biological replicates). One way RM ANOVA revealed no significant difference among groups. $F(2,10)=1.336$, $P=0.3059$.

d, Total eIF4E levels are significantly reduced by srIkBa, an effect reversed by co-expression of Smad7 and SOCS1 (n = 6 biological replicates). Statistical significance was determined by one-way RM ANOVA and Holm-Šidák's multiple comparisons test to compare all groups.

saRNA constructs are described in [Fig. 4a](#). Connecting lines indicate responses from the same biological replicate. For all statistical reporting, * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$. Mock transfection data used for normalization are the same as in [Fig. 3](#).

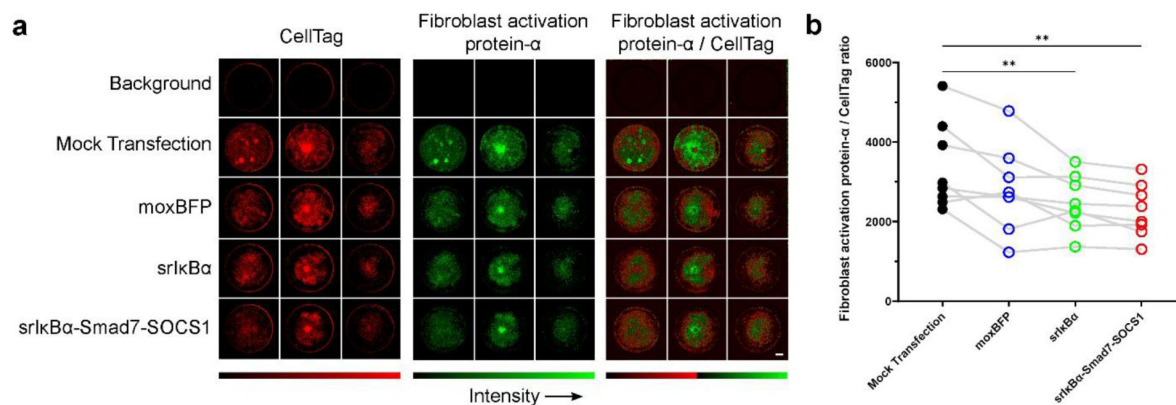


Fig. 7.

Prolonged transfection with srIkBα or srIkBα-Smad7-SOCS1 significantly reduces basal fibroblast activation factor-α (FAP-α) levels.

a, Representative in-cell western images showing FAP-α expression. Columns show different biological replicates, and rows show different treatments. The montage on the right shows FAP-α signal normalized to CellTag signal (FAP-α/CellTag).

b, In-cell western assay of FAP-α expression (n = 8 biological replicates). Both srIkBα and srIkBα-Smad7-SOCS1 significantly reduce FAP-α levels compared to mock transfection, while moxBFP does not differ significantly from mock. Statistical significance was determined by one-way RM ANOVA with Greenhouse–Geisser correction and Dunnett's multiple comparisons test to compare groups to mock transfection. Connecting lines indicate responses from the same biological replicate. **P < 0.01.

External control of transgene expression using a small-molecule antiviral

The srIkBa-Smad7-SOCS1 construct, which includes E3-NSs-L*, provides broad suppression of dsRNA sensing and inflammatory signaling pathways. This construct mitigates saRNA-induced cytokine release (Figs. 4b, S4a-m) and counters saRNA- and srIkBa-induced reductions in cell viability (Figs. 5c, S5a-c), enabling sustained cap-dependent transgene expression (Fig. 5e) without inducing fibroblast activation (Fig. 7b) or causing srIkBa-associated reductions in eIF4E levels (Fig. 6d). Together, these attributes position the srIkBa-Smad7-SOCS1 construct as a promising candidate for transient gene therapy applications.

To further enhance the therapeutic potential of this construct, we investigated the possibility of using a small-molecule drug to halt transgene expression. Achieving temporal control of transgene expression would enhance safety and flexibility in therapeutic contexts. The saRNA constructs in this study rely on the VEEV RdRp for self-amplification. VEEV is a mosquito-borne alphavirus capable of causing flu-like illness in humans, with the potential to progress to fatal encephalitis.⁷² Due to its pathogenicity, significant efforts have focused on developing small-molecule inhibitors of VEEV. One such compound is ML336, a potent inhibitor of the VEEV RdRp.⁷³

To test whether ML336 could inhibit transgene expression of the srIkBa-Smad7-SOCS1 construct, ML336 or vehicle was added to the culture medium one day post-transfection. EGFP and mScarlet3 fluorescence were monitored over 13 days (Fig. 8a), followed by calcein AM staining to assess cell viability (Fig. 8b).

ML336 treatment significantly reduced EGFP and mScarlet3 expression over time (Figs. 8c, 8d), demonstrating its efficacy in controlling saRNA transgene expression. While vehicle-treated cultures exhibited lower viability compared to mock-transfected cells—possibly due to reduced proliferation over the experiment's duration—prolonged ML336 treatment did not reduce viability relative to mock-transfected controls (Fig. 8e).

Discussion

saRNA-induced translation shutdown is thought to be mediated by phosphorylation of eIF2α.^{22,74,75} During saRNA replication, dsRNA replicative intermediates activate PKR, leading to eIF2α phosphorylation and translation inhibition.^{7,22} Our study is consistent with this mechanism and additionally demonstrates that saRNA induces a reduction in eIF4E phosphorylation, a phenomenon previously observed with alphavirus replicons, although the underlying mechanism remains unclear.^{76,77} Similar to eIF2α phosphorylation,⁷⁸ a reduction in eIF4E phosphorylation decreases global cap-dependent translation initiation,⁷⁹ promoting

cap-independent translation.⁸⁰ These combined effects—phosphorylation of eIF2α and reduced phosphorylation of eIF4E—may explain why IRES-driven, cap-independent transgenes often achieve higher expression compared to those relying on cap-dependent expression when using native saRNA.^{26,81}

To counteract saRNA-induced translation shutdown, we designed saRNA constructs that express inhibitors targeting dsRNA-sensing pathways using IRES-mediated translation. During translation shutdown, global mRNA translation decreases, freeing ribosomes that would otherwise be used for translating capped mRNAs, thereby enhancing cap-independent translation.⁸² By leveraging this mechanism, IRES-mediated expression of dsRNA-sensing pathway inhibitors ensures their levels

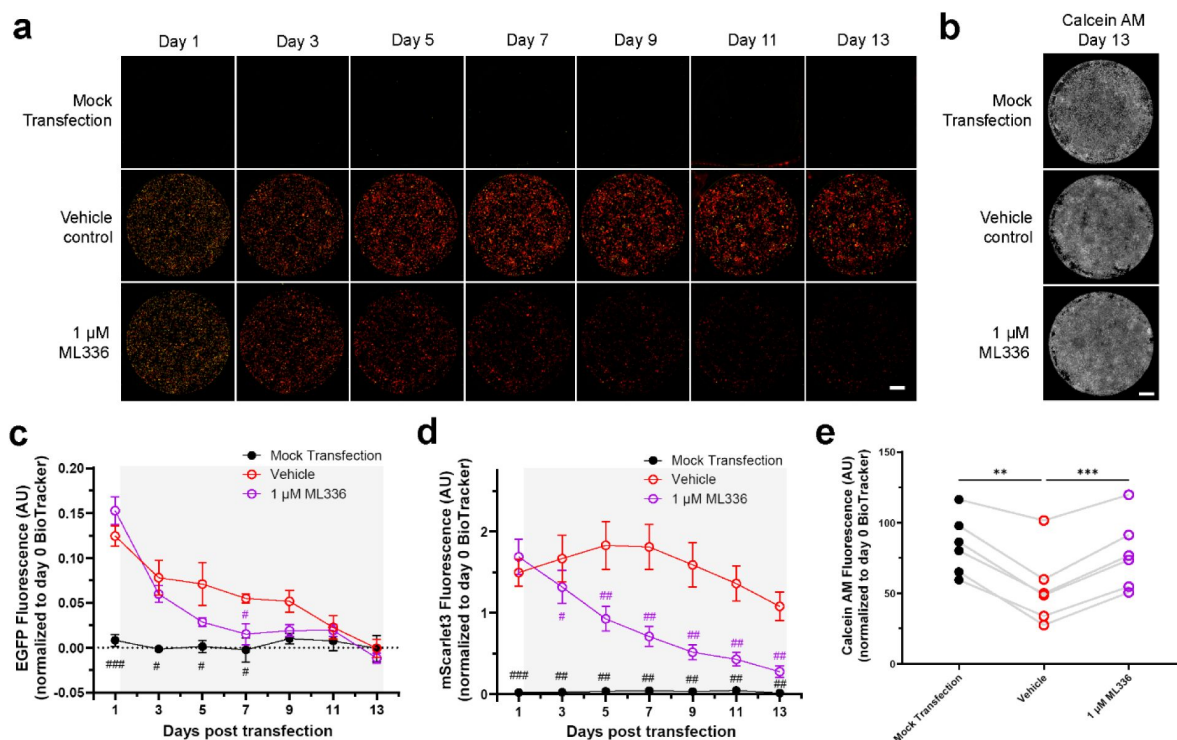


Fig. 8.

ML336 enables external control of transgene expression from the srIkBa-Smad7-SOCS1 construct.

a, Representative images of EGFP (green) and mScarlet3 (red) expression over 13 days in FLS transfected with srIkBa-smad7-SOCS1. Cultures were treated with vehicle or 1 μ M ML336, starting 1 day post-transfection. Scale bar = 5 mm.

b, Representative images of calcein AM staining on day 13 post-transfection. Scale bar = 5 mm.

c, Quantification of EGFP fluorescence intensity in srIkBa-Smad7-SOCS1-transfected FLS ($n = 6$ biological replicates). Cultures were treated with vehicle or 1 μ M ML336 (treatment period indicated by shading). EGFP fluorescence was significantly lower in ML336-treated cultures compared to vehicle-treated cultures on day 7. Statistical significance relative to vehicle-treated cells was determined by two-way RM ANOVA with Greenhouse-Geisser correction and Dunnett's multiple comparisons test. # $P < 0.05$, ### $P < 0.01$.

d, Quantification of mScarlet3 fluorescence intensity in srIkBa-Smad7-SOCS1-transfected FLS ($n = 6$ biological replicates). Cultures were treated with vehicle or 1 μ M ML336 (treatment period indicated by shading). mScarlet3 fluorescence was significantly lower in ML336-treated cultures compared to vehicle-treated cultures beginning on day 3. Statistical significance relative to vehicle-treated cells was determined by two-way RM ANOVA with Greenhouse-Geisser correction and Dunnett's multiple comparisons test. # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$.

e, Quantification of calcein AM staining on day 13 post-transfection, following 12 days of vehicle or ML336 treatment ($n = 6$ biological replicates). Cultures treated with ML336 showed no significant difference in calcein AM signal compared to mock transfection, while vehicle-treated cultures exhibited significantly lower calcein AM signals compared to both mock-transfected and ML336-treated cultures. Connecting lines indicate responses from the same biological replicate. Statistical significance was determined by one-way RM ANOVA with Greenhouse-Geisser correction and Tukey's multiple comparisons test. ** $P < 0.01$, *** $P < 0.001$.

Data were normalized to BioTracker intensity on day 0 and are presented as mean $\pm$ SEM.

increase as translation shutdown intensifies. This approach effectively inhibits dsRNA sensing and alleviates translation shutdown. In contrast, encoding these inhibitors via cap-dependent translation is expected to be less effective, as translation shutdown would lead to reduced inhibitor expression.

In this study, we used vaccinia virus E3 to prevent saRNA-induced translation shutdown, a role that E3 has previously been used to fulfill.^{22,23} E3 achieves this by sequestering dsRNA intermediates from sensors like PKR⁸³ and by directly inhibiting PKR activation.⁸⁴ These dual actions prevent eIF2 α phosphorylation, alleviating translation inhibition and enhancing transgene expression.^{22,23} However, our findings suggest that this mechanism is incomplete. While E3 prevents saRNA-induced eIF2 α phosphorylation, its ability to enhance transgene expression also depends on RNase L activity. By alleviating translation inhibition, E3 facilitates unrestrained RdRp production, driving robust saRNA replication and dsRNA synthesis. This dsRNA activates RNase L, which degrades mRNA⁵⁰ and inhibits nuclear mRNA export.⁵¹ The resulting depletion of cytoplasmic mRNA⁵² reduces competition for ribosomes, enabling enhanced translation of saRNA transcripts. These transcripts are shielded from RNase L degradation due to their localization within plasma membrane-associated micro-compartments called spherules.³ Consistent with this model, inhibiting RNase L with L* in the E3-NSs-L* construct reduced transgene expression, likely due to restored ribosomal competition from intact host mRNA.

While preventing eIF2 α phosphorylation and activating RNase L drive high levels of saRNA transgene expression, this combination also causes significant cytotoxicity with the E3 construct. This dual-edged effect potentially makes the E3 construct suited for immunotherapy applications, such as vaccines and cancer therapy, where robust transgene expression and immunogenic cell death can promote strong immune responses.^{85,86} In contrast, non-immunotherapy applications require strategies that prioritize cell viability and minimize tissue damage over maximizing transgene expression.

Alphavirus vectors, including the VEEV replicon used in this study, are known to induce cytotoxicity in mammalian cells.^{6,10,12–16} This cytotoxicity arises from dsRNA intermediates, which activate sensors such as PKR, RIG-I, MDA-5, and OAS,^{5,7,22,87} despite partial shielding within spherules.^{3,74} Studies in HeLa cells using microinjected dsRNA show that PKR is essential for cytosolic dsRNA-induced apoptosis, while RIG-I and MDA-5 are dispensable.⁸⁸ Numerous studies further underscore the central role of PKR in mediating apoptosis in response to dsRNA,^{89–91} and have shown that PKR can mediate dsRNA-induced apoptosis even in the absence of eIF2 α phosphorylation.⁹² OAS also contributes to dsRNA-induced apoptosis by producing 2'-5'-oligoadenylates upon binding dsRNA, which activate RNase L, leading to cleavage of single-stranded RNA.⁵² This disrupts cellular translation as cleavage of actively translated mRNA causes ribosome stalling. Ribosomes encountering truncated 3' ends of mRNAs are unable to dissociate, resulting in collisions,^{93,94} triggering the ribotoxic stress response,⁹⁵ which can lead to apoptosis.^{93,96} These findings from the literature identify PKR and OAS/RNase L as key contributors to saRNA-associated cytotoxicity and highlight them as valuable targets for mitigation.

While E3 relieves saRNA-induced eIF2 α phosphorylation,^{22,23} the E3 construct also compromised RNA integrity, suggesting activation of the OAS/RNase L pathway. Furthermore, PKR, an IFN-inducible gene,⁹⁷ was upregulated by the E3 construct, potentially allowing partial evasion of E3-mediated inhibition. Thus, vaccinia virus E3 alone did not fully address the cytotoxic actions of dsRNA replicative intermediates produced during saRNA replication. To further suppress PKR and OAS/RNase L pathways, we used Toscana virus NSs³⁹ and Theiler's virus L*⁴⁰ respectively. While this strategy resulted in lower transgene expression—likely due to the protective actions of RNase L inhibition on cellular mRNA integrity—it effectively mitigated the intrinsic mechanisms of saRNA cytotoxicity. However, the inclusion of multiple viral proteins introduces additional challenges. *In vivo*, saRNA constructs must contend not only with intrinsic

cytotoxic mechanisms but also with adaptive immune responses leading to extrinsic cytotoxicity. The co-expression of several viral proteins, along with the VEEV RdRp, significantly increases the likelihood of viral peptides being presented on major histocompatibility complex class I molecules to cytotoxic T lymphocytes. This is expected to activate adaptive immune responses, which could culminate in the elimination of saRNA-transfected cells. Addressing both intrinsic and immune cell-mediated cytotoxicity will be critical to achieving the long-term viability of saRNA-transfected cells and unlocking the platform's full therapeutic potential.

While we found that inhibiting dsRNA-sensing pathways was an effective means to reduce saRNA-induced cytotoxicity, it was not effective at preventing saRNA-induced cytokine responses. This agrees with prior reports showing that dsRNA activates NF- κ B even in the absence of PKR and RNase L,⁹⁸ suggesting additional mechanisms contribute to NF- κ B activation. One such mechanism may involve eIF4E, a key regulator of translation initiation, whose activity is governed by its phosphorylation status and serves as a rate-limiting step in cap-dependent protein synthesis.⁴⁹ In this study, we found that saRNA transfection induces a reduction in eIF4E phosphorylation. Reduced eIF4E phosphorylation has been shown to decrease the levels of the short-lived I κ B α protein,⁹⁹ an inhibitor of NF- κ B that sequesters it in the cytoplasm and prevents nuclear translocation. Decreased I κ B α levels thus allow NF- κ B to translocate to the nucleus, initiating cytokine expression and inflammatory responses. To address this, we expressed srI κ B α using cap-independent translation to produce a stable cytoplasmic pool of I κ B α , thereby preventing NF- κ B activity. While this strategy effectively reduced saRNA-induced cytokine responses, it also resulted in a reduction in cell viability.

The NF- κ B pathway is a key regulator of inflammatory cytokine production, making it an important target for suppressing cytokine responses.¹⁰⁰ However, NF- κ B also has pro-survival functions as it promotes the transcription of anti-apoptotic genes—though its exact effects depend on tissue type and biological context.¹⁰¹ In FLS, overexpression of srI κ B α reduces inflammatory cytokine production but increases sensitivity to TNF- α -induced apoptosis,^{102,103} consistent with our findings that saRNA encoding srI κ B α lowered both anti-viral cytokine responses and cell viability. This phenomenon is not unique to FLS; inhibition of NF- κ B signaling with srI κ B α has also been shown to render liver hepatocytes more susceptible to pro-apoptotic stimuli.¹⁰⁴ This raises a key safety consideration, as saRNA delivery systems such as lipid nanoparticles tend to accumulate in the liver.¹⁰⁵ Therefore, when designing saRNA constructs for sustained, non-immunostimulatory transgene expression in contexts where apoptosis is undesirable, the use of srI κ B α to suppress cytokine responses may require co-expression of anti-apoptotic proteins to offset the loss of NF- κ B-mediated survival signaling.

In this study, we show that co-expressing Smad7 and SOCS1 effectively counteracted srI κ B α -induced cell death in FLS. While we did not assess the individual contributions of each protein, both are known to have anti-apoptotic properties in certain contexts. Smad7 can inhibit TGF- β -induced apoptosis,^{106,107} while SOCS1 can suppress apoptosis triggered by IFN- α , IFN- β ,¹⁰⁸ IFN- γ ,¹⁰⁹ and TNF- α .^{110–112} These anti-apoptotic effects likely enable Smad7 and SOCS1 to mitigate srI κ B α -induced reductions in FLS cell viability.

We also found that co-expression of Smad7 and SOCS1 significantly enhanced cap-dependent transgene expression. Although we did not distinguish the contributions of each protein to this effect, it is likely mediated by SOCS1. Previous studies have shown that ruxolitinib, a small-molecule inhibitor of Janus kinase (JAK) 1 and JAK2,¹¹³ increases saRNA transgene expression when added to saRNA polyplexes.¹⁸ SOCS1 is an endogenous inhibitor of JAK1 and JAK2,¹¹⁴ therefore, SOCS1 offers a genetically encoded alternative to the small-molecule approach of ruxolitinib for enhancing saRNA expression.

Furthermore, this study presents a straightforward method for controlling saRNA activity. ML336, a VEEV RdRp inhibitor, blocks the synthesis of positive-sense genomic, negative-sense template, and subgenomic RNAs of VEEV without affecting host cellular RNA transcription.¹¹⁵ It has an *in vitro* effective concentration 50 (EC₅₀) of 0.02–0.04 μM, and a 50 mg/kg dose offers 100% protection in a lethal VEEV mouse infection model, with no observed toxicity.¹¹⁶ Thus, ML336 and other RdRp inhibitors could provide a practical means to deactivate saRNA—either to terminate transgene expression once therapeutic objectives are met or as a safety measure to reduce the risks associated with prolonged innate immune suppression and other potential side effects, including adaptive immune responses to the VEEV replicon or viral innate immune inhibitory proteins.

Preclinical and clinical data show that saRNA vaccines often trigger an intense innate immune response that can hinder antigen expression.^{8,18,22} Additionally, saRNA vaccines can induce significant inflammation,^{117,118} which is increasingly recognized as problematic, as common adverse events include pain, headache, tenderness, arthralgia, fever, and chills.^{119,120} The approaches described in this study, which reduces the inherent immunostimulatory effects of saRNA replication, could potentially be applied to future saRNA vaccine designs to improve antigen expression and reduce reactogenicity.

The approach in this study also underscores the versatility of the saRNA platform, which can efficiently integrate multiple transgenes and genetic elements. Our largest construct included the VEEV RdRp, two IRES sequences, five 2A peptides, and eight transgenes within a 16.5-kilobase saRNA. This ability to encode numerous transgenes without apparent size constraints is a notable yet underutilized strength of the saRNA platform.

Leveraging this capacity for multiple transgenes, we encoded diverse inhibitors of dsRNA-sensing and inflammatory signaling pathways directly within saRNA achieving durable and controllable transgene expression with minimal cytotoxicity and immunostimulation. This approach eliminates the need for exogenous immune suppressants, a common requirement in other studies utilizing saRNA for non-immunotherapy applications. For instance, therapeutic antibody production, cellular reprogramming, and retinal cell transfection with saRNA have relied on interferon-α/β receptor-1 blocking antibodies or the vaccinia virus B18R interferon decoy receptor to suppress innate immune responses and achieve effective transgene expression.^{9,21,26,27} Similarly, corticosteroids have been used to enhance antigen expression by reducing saRNA-induced innate immune activation.²⁰ Here, we demonstrate that encoding broad-spectrum innate immune suppression directly within the saRNA obviates the need for additional immune-modulatory treatments, thereby reducing treatment complexity. Moreover, this strategy localizes immune suppression to transfected cells, which should minimize the risks of off-target effects associated with systemic immune suppression.

Expressing multiple transgenes may also be a useful strategy for developing novel therapeutics. The NF-κB, TGF-β, and JAK/signal transducer and activator of transcription (STAT) pathways are central to the progression of osteoarthritis.^{121–123} The srIkBa-Smad7-SOCS1 saRNA construct provides a novel strategy to simultaneously target these pathways: srIkBa inhibits NF-κB, Smad7 blocks TGF-β, and SOCS1 suppresses the JAK/STAT pathway. Each of these proteins has been independently studied as a treatment for osteoarthritis,^{124–126} and their combination within a single construct could represent a synergistic strategy to enhance therapeutic efficacy. While arthritogenic alphaviruses such as chikungunya virus and Ross river virus can drive inflammatory responses in FLS, including upregulation of secreted proteolytic molecules such as matrix metalloproteases,^{127,128} the lack of FAP-α upregulation in transfected FLS provides some evidence that saRNA transfection does not trigger comparable responses. Moreover, the reduction in basal FAP-α levels observed with the srIkBa and srIkBa-Smad7-SOCS1 constructs

raises the possibility that saRNA could serve as a tool for suppressing fibroblast activation in inflammatory settings. This approach could hold therapeutic potential in arthritis treatment,¹²⁹ warranting further investigation in disease models.

In addition to expressing therapeutic transgenes, saRNA holds potential for inducing therapeutic effects by modulating translational control. While the strategy outlined in this report mitigated eIF2 α phosphorylation, it was not effective at preventing saRNA-induced reductions in eIF4E phosphorylation. *In vitro* directed evolution of the RdRp replicon^{6,24,130} may offer a potential solution to this challenge.

Furthermore, we found that srIkBa expression decreased eIF4E levels, consistent with previous findings.¹³¹ While reducing eIF4E activity through saRNA or srIkBa expression could lower transgene expression, it may also have therapeutic value, as lower eIF4E activity is associated with decreased pain sensitivity,¹³² inhibition of tumorigenesis,¹³³ improvement of neurodevelopmental social deficits,¹³⁴ and increased longevity.¹³⁵ We also observed that srIkBa expression reduced phosphorylation of eIF2 α , while the E3-NSs-L* construct reduced PKR levels. Given that eIF2 α phosphorylation activates the integrated stress response, a pathway often driven by PKR in age-related diseases like neurodegeneration,¹³⁶ saRNA may represent a novel approach to modulate stress responses in these conditions.

Our findings present a fully saRNA-based strategy to address a key limitation of saRNA: replication driven innate immune stimulation. By inhibiting multiple innate immune pathways through proteins expressed via cap-independent translation, we mitigate the intrinsic cytotoxicity of saRNA. This approach enables sustained transgene expression with minimal cytotoxicity and antiviral cytokine release, all without the need for exogenous immunosuppressants. Given the established success of saRNA *in vivo*—including its application in preclinical models^{6,18,22,26} and approved human vaccines^{120,137}—we anticipate that this strategy will perform effectively *in vivo*, which future studies will address. This work advances the therapeutic potential of saRNA for non-immunotherapy applications, including protein replacement therapy, therapeutic antibody production, genome editing, cellular reprogramming, and immune cell engineering, where prolonged innate immune stimulation is undesirable. Importantly, the ability to externally regulate transgene expression offers precise control and serves as a critical safety mechanism for future clinical applications.

Materials & methods

Experimental Design

This study aimed to develop and evaluate saRNA constructs that modulate innate immune responses, specifically targeting dsRNA sensing and inflammatory signaling pathways, to reduce saRNA-induced cytotoxicity, cytokine expression, and achieve sustained transgene expression. To this end, we engineered saRNA constructs that co-express multiple viral innate immune inhibitors and cellular inhibitors of inflammatory signaling. The constructs were designed to enable monitoring of cap-dependent and cap-independent translation using distinct fluorescent proteins.

The experimental approach included cloning the saRNA constructs using molecular biology techniques, followed by *in vitro* transcription and transfection into mouse primary FLS. Transfected cells were analyzed using fluorescence-based microplate assays to assess cell number, cell viability, phosphatidylserine exposure, and fluorescent protein expression. Further evaluations of translational control, FLS activation markers, RNA integrity, and cytokine secretion were conducted using in-cell western assays, capillary electrophoresis, and bead-based immunoassays. Finally, the ability to control saRNA transgene expression was explored using a small-molecule inhibitor targeting the VEEV RdRp.

Plasmid cloning and construct design

The commercially available TagGFP2 E3L Simplicon vector (SCR725, Merck) served as the saRNA backbone in this study. After removing the TagGFP2-IRES-E3 sequence, various constructs were generated by combining different elements using restriction digests (FastDigest, Thermo Scientific), overlap extension PCR (Phusion, Thermo Scientific), and HiFi assembly (NEBuilder, New England Biolabs).

Genetic elements were sourced as follows: mScarlet3 from pDx_mScarlet3 (a gift from Dorus Gadella [Addgene plasmid #189754]),¹³⁸ IRES-EGFP from pIRES2-EGFP (Clontech), and IRES-E3 and IRES-PuroR (puromycin resistance) from the original TagGFP2 E3L Simplicon vector.

Custom gene synthesis (GeneArt, Thermo Scientific) was used to produce IRES-moxBFP, IRES-E3-T2A-NSs-P2A-L*-E2A-mEGFP, and IRES-srIkBa-P2A-Smad7-T2A-SOCS1 sequences. The IRES sequence was derived from pIRES2-EGFP (Clontech). Other sequences were derived from public databases: moxBFP (FPbase ID: SSTDU),¹³⁹ mEGFP (FPbase ID: QKFJN), E3 (UniProt: P21605-1), NSs (UniProt: P21699), L* (UniProt: P0DJX4), Smad7 (UniProt: O35253-1), and SOCS1 (UniProt: O35716). srIkBa was engineered from IkBa (UniProt: Q9Z1E3) by substituting serines 32 and 36 with alanines. All synthesized sequences were codon-optimized for mouse expression (GenSmart, GenScript).

Plasmids were cloned in DH5a competent *Escherichia coli* (High Efficiency, New England Biolabs) and purified by maxiprep (PureLink HiPure, Invitrogen). All plasmid sequences were verified using nanopore whole plasmid sequencing (Plasmidsaurus).

RNA synthesis

Plasmids were linearized using XbaI (FastDigest, Thermo Scientific) at 37°C for 3 hours. The linear plasmids were purified by phenol-chloroform extraction followed by sodium acetate ethanol precipitation. Uncapped RNA was synthesized *in vitro* using the T7 RiboMAX Large Scale RNA Production kit (Promega) at 37°C for 2 hours. After purification by ammonium acetate precipitation, the RNA was denatured at 65°C for 5 minutes. Cap-1 structures were then generated using the Vaccinia Capping System in conjunction with mRNA cap 2'-O-methyltransferase (both from New England Biolabs) for 45 minutes at 37°C. Following another ammonium acetate purification, the RNA was treated with Antarctic phosphatase (New England Biolabs) for 30 minutes at 37°C. After a final ammonium acetate purification step, the RNA was resuspended in THE RNA Storage Solution (Invitrogen), aliquoted, and stored at -80°C until use.

Animals

All mice used in this study were 5- to 12-week-old wildtype C57BL/6J mice (Bicester, UK). Both male and female mice were included, though sex differences in response to saRNA transfection were not analyzed. Mice were housed in groups of up to five in a temperature-controlled room (21°C), with bedding, a red shelter, and enrichment materials provided. They were maintained on a 12-hour light/dark cycle, with food and water available *ad libitum*. Mice were euthanized by exposure to CO₂ gas in a rising concentration, and death confirmed by dislocation of the atlanto-occipital joint by applying pressure to the side of the neck using thumb and forefinger, procedures in line with the Animals (Scientific Procedures) Act 1986 Amendment Regulations 2012.

Mouse primary FLS culture

We cultured FLS from mouse knee joints following established protocols,¹⁴⁰ with slight modifications. In brief, knee joints were exposed by cutting through the overlying skin with dissecting scissors. The patellar tendon was then grasped near the patella with Dumont tweezers,

and spring scissors were used to sever the patellar tendon along with surrounding tissues. Finally, the patella was excised by cutting the quadriceps tendon, and excess muscle and tendons were carefully trimmed away.

Both patellae from each animal were collected into a microcentrifuge tube containing ice-cold sterile phosphate buffered saline (PBS), then transferred to a 24-well tissue culture plate (Costar) with FLS culture media. The media consisted of DMEM/F-12 (Invitrogen) supplemented with 25% fetal bovine serum (Sigma), 1X GlutaMAX (Gibco), and 100 µg/ml Primocin (InvivoGen).

The patellae were maintained in a humidified incubator at 37°C with 5% CO₂, with media changes every 1-2 days. FLS outgrowth was observed, and after approximately 10 days, when cells reached 70% confluency, the patellae were transferred to new wells for further FLS collection. Cells attached to the original wells were passaged using 0.1% trypsin-EDTA (Sigma) into a single well of a 6-well plate (Costar). Cells were later expanded into T25 flasks (BioLite, Thermo Scientific or CELLSTAR, Greiner Bio-One) and passaged upon reaching confluency. After the second passage, Versene solution (Gibco) was used for cell dissociation.

FLS were used between passages 3 and 8. In most experiments, cells from the same animal were divided and plated onto 6-well or 24-well plates, allowing for parallel manipulations with matched conditions. Occasionally, when cell number was low, cells from different animals were pooled before division. Once plated for experiments, media was changed every 2-3 days.

Immunocytochemistry and confocal microscopy

Cells were plated on poly-D-lysine-coated glass-bottom 35 mm dishes (P35GC-1.5-14-C, MatTek) and fixed with 4% paraformaldehyde for 10 minutes at room temperature without permeabilization. After fixation, cells were washed twice with PBS, followed by blocking with 10% normal goat serum in PBS for 1 hour at room temperature.

Cells were then incubated overnight at 4°C in blocking buffer, either without primary antibody (no primary control) or with cadherin-11 (extracellular) rabbit polyclonal antibody (1:200, DF3523, Affinity Biosciences). After three washes with PBS, cells were incubated for 1 hour at room temperature in blocking buffer with 1 µg/ml Hoechst 33342, BioTracker NIR680 (1:2000, Merck), and goat anti-rabbit Alexa Fluor 488 secondary antibody (1:500, Invitrogen).

After four additional PBS washes, cells were imaged in PBS using a Leica Stellaris 5 confocal microscope equipped with a 40X oil immersion objective. Tile-scanned images were stitched using Las X microscopy software (Leica) to generate high-resolution panoramic images.

BioTracker staining

BioTracker NIR680 was diluted 1:2000 in unsupplemented DMEM/F-12. T25 flasks containing FLS were washed once with HBSS (with calcium and magnesium, Gibco), then incubated with the diluted dye for 30 minutes at 37°C. After incubation, the flasks were washed three times with FLS media, with each wash lasting 10 minutes at 37°C. The FLS were then dissociated using Versene and plated onto 6- or 24-well plates.

FLS transfection

For transfection in 6-well plates, the medium was removed and replaced with 1 ml of Opti-MEM I (Gibco). In a microcentrifuge tube, 500 ng of saRNA was diluted in 200 µl of Opti-MEM I. In a separate tube, 3 µl of Lipofectamine MessengerMAX (Invitrogen) was diluted in 100 µl of Opti-MEM I. After gently mixing, the complexes were incubated at room temperature for 5 minutes,

then added dropwise to the cells. The cells were incubated with the complexes for 2 hours at 37°C, after which the media was removed and replaced with fresh FLS media. For transfection in 24-well plates, all volumes and amounts of saRNA were reduced fivefold.

Microplate imaging

Unless otherwise specified, black glass-bottom 6-well plates (P06-1.5H-N, Cellvis) were used for microplate imaging. Prior to imaging, the FLS media was replaced with Live Cell Imaging Solution (Invitrogen). Imaging was performed using the Odyssey M laser scanner (LI-COR), using LI-COR acquisition software with a plate offset of +1.45 mm and 100 µm resolution.

EGFP, mScarlet3, and BioTracker NIR680 were imaged using the 488, 520, and 700 channels, respectively. Spectral cross-excitation between EGFP and mScarlet3 in the 488 and 520 channels was corrected through linear unmixing analysis, as described below. Fluorescence intensity was quantified in ImageJ by measuring the integrated density within equal-area regions of interest for each well.

Fluorescence values (EGFP, mScarlet3, and BioTracker) were normalized by the day 0 BioTracker signal measured before transfection to account for variations in starting cell number.

Linear unmixing analysis

saRNA constructs encoding individual fluorescent proteins (moxBFP, EGFP, or mScarlet3) were designed to assess cross-excitation among the fluorescent proteins used in this study. tSA201 cells (96121229, ECACC) were seeded in black glass-bottom 6-well plates coated with poly-D-lysine (Sigma) and transfected with the saRNA constructs. Cells were imaged the following day using an Odyssey M laser scanner, with fluorescence captured through the 488 and 520 channels. Expression of moxBFP was undetectable in both the 488 and 520 channels, and therefore it was excluded from further analysis.

Bleed-through of EGFP into the 520 channel and mScarlet3 into the 488 channel was quantified using ImageJ, with cross-excitation determined to be 11.32% and 0.94%, respectively. The emission signals were assumed to be linearly proportional to the sum of the intensities of each fluorophore, and the unmixed EGFP and mScarlet3 signals were calculated using Python.

Annexin V assay

FLS were stained with BioTracker NIR680 and plated on black glass-bottom 24-well plates (Sensoplate, Greiner Bio-One). Cells were transfected with saRNA or treated with 0.5 µM staurosporine (Cayman Chemical) as a positive control.

Annexin V-CF800 conjugate (Biotium) was diluted to 250 ng/ml in Live Cell Imaging Solution (Invitrogen). On each day of the assay, wells were washed with Live Cell Imaging Solution and replaced with diluted Annexin V-CF800 solution. Plates were incubated at 37°C for 15 minutes. Following incubation, plates were washed 3 times with Live Cell Imaging Solution before imaging the 700 and 800 channels with an Odyssey M imager set to +1.45 mm image offset and 100 µm resolution.

Image analysis was performed using ImageJ. To correct for unidirectional spectral bleed-through of BioTracker NIR680 into the 800-channel, subtractive compensation was applied by dividing the 700-channel image by a factor of 800 and then subtracting it from the 800-channel image. Due to the presence of a small number of high-intensity speckles in the 800-channel image, area rather than fluorescence intensity was quantified. The display range of the 800-channel image was set between 0.25 and 2.5 and the image was thresholded using the method of Li¹⁴¹. The area of

thresholded pixels within each well was then quantified and adjusted based on the BioTracker signal before transfection to account for variations in cell number. Data was then normalized to the average of the mock transfection condition.

Calcein AM staining

Calcein AM (Invitrogen) staining was performed according to the manufacturer's protocol. Briefly, cells were washed with HBSS and incubated with 2 μ M calcein AM (diluted in HBSS) at 37°C for 30 minutes. Following three washes, cells were imaged in Live Cell Imaging Solution, and the 488-channel image was captured using an Odyssey M imager.

Although calcein AM and EGFP share overlapping spectra, calcein AM fluorescence was typically much higher than EGFP, making EGFP's contribution to the measured Calcein AM signal negligible in most cases. However, when the E3 construct was used, EGFP expression was sufficient to interfere with the signal. To address this, in experiments that included use of the E3 construct, a 488-channel image was captured prior to calcein AM application and subtracted from the post-application image to accurately measure calcein AM fluorescence.

Fluorescence intensity was quantified using ImageJ. In experiments where BioTracker was used, fluorescence values were corrected by the pre-transfection BioTracker signal to account for differences in starting cell number.

In-cell western assay

Cells were plated on black, glass-bottom 24-well plates. Alongside mock and saRNA transfections, one well was reserved for background subtraction, which received no treatment. Unless specified otherwise, experiments were conducted 2 days post-transfection. Cells were fixed with 4% paraformaldehyde for 10 minutes at room temperature, followed by two washes with tris-buffered saline (TBS). Permeabilization was then performed using 0.1% Triton X-100 in TBS for 10 minutes, followed by two additional TBS washes. After permeabilization, cells were blocked with Intercept TBS Blocking Buffer (LI-COR) for 1 hour at room temperature with gentle agitation.

Primary antibodies were diluted in Intercept Blocking Buffer and incubated with the cells overnight at 4°C. The background subtraction well was incubated with blocking buffer alone, without primary antibodies.

The following primary antibodies were used: Phospho-eIF2 α (Ser51) (D9G8) XP rabbit monoclonal (1:200, #3398, Cell Signaling Technology), eIF2 α (L57A5) mouse monoclonal (1:200, #2103, Cell Signaling Technology), PKR rabbit polyclonal (1:500, #18244-1-AP, Proteintech), Phospho-eIF4E (S209) rabbit monoclonal (1:200, #ab76256, Abcam), eIF4E mouse monoclonal (5D11) (1:200, #MA1-089, Invitrogen), and fibroblast activation protein (73.3) mouse monoclonal (20 μ g/ml, #BE0374, InVivoMAB). Except for the fibroblast activation protein antibody, which was conjugated to DyLight 800 using the manufacturer's DyLight Antibody Labeling Kit (#53062, Thermo Scientific), all primary antibodies were unconjugated.

After washing the wells three times with TBS, cells were incubated for 1 hour at room temperature with the appropriate secondary antibodies or normalization stain, with gentle agitation. The secondary antibodies used were goat anti-mouse IRDye 800CW (1:800), donkey anti-rabbit IRDye 800CW (1:800), donkey anti-mouse IRDye 680RD (1:800), and CellTag 700 (1:500), all from LI-COR. The background subtraction well received secondary antibody but no CellTag. Following four additional washes with TBS, the plate was inverted and gently tapped on absorbent paper to remove excess liquid. The plate was then imaged using the Odyssey M imager with LI-COR Acquisition software using a plate offset of +1.45 and 100 μ m resolution. Signal quantification was carried out using Empiria Studio software (LI-COR).

rRNA integrity analysis

FLS were plated in 6-well plates and either transfected with saRNA or subjected to mock transfection. After 26-30 hours (allowing for saRNA replication but before substantial cell loss), cells were harvested by scraping in 350 μ l RLT buffer (QIAGEN) supplemented with 10 μ l/ml β -mercaptoethanol (Sigma). Cell lysates were homogenized using QIAshredder columns (QIAGEN), and total RNA was extracted using the RNeasy Mini Kit (QIAGEN) according to the manufacturer's instructions. RNA quantity and quality were initially assessed using a N60 nanophotometer (Implen). Samples were aliquoted and stored at -80°C until further analysis. For detailed quality assessment, samples were transported on dry ice and analyzed using the RNA Pico Kit on the 2100 Bioanalyzer system (Agilent), performed by Cambridge Genomic Services. RNA Integrity Number (RIN) values were determined for each sample.

Bead-based immunoassay

FLS were stained with BioTracker and seeded onto 6-well plates. After transfection, the saRNA-lipofectamine complexes were replaced with 2 ml of fresh FLS media. Two days later, supernatants were collected and stored at -80°C until analysis. On the day of analysis, the supernatant was thawed on ice, and levels of 13 innate immune-related cytokines (IFN- γ , CXCL1, TNF- α , MCP-1, IL-12p70, CCL5, IL-1 β , CXCL10, GM-CSF, IL-10, IFN- β , IFN- α , and IL-6) were measured using the LEGENDplex Mouse Anti-Virus Response Panel (740621, BioLegend) following the manufacturer's protocol. Samples were run in duplicate. A custom 3D-printed PETG vacuum manifold (designed with Fusion 360) was used for vacuum filtration, and beads were analyzed using a CytoPLEX LX flow cytometer (Beckman Coulter). Data analysis was performed with the LEGENDplex Qognit Data Analysis Suite (BioLegend). To account for differences in cell number between wells, cytokine levels were corrected based on the BioTracker signal before transfection.

ML336 experiments

ML336 (Cayman Chemical) was dissolved in DMSO (Sigma) at a stock concentration of 10 mM, and aliquots were stored at -20°C. For treatments, ML336 was added to cell cultures at a final concentration of 1 μ M, while vehicle control cultures received 0.01% DMSO.

Statistical Analysis

All statistical analyses were conducted using GraphPad Prism 9, with specific tests detailed in the corresponding figure legends. All statistical tests were two-sided tests.

Data availability statement

The code for the unmixing analysis is available at https://github.com/lariferg/spectral_unmixing.

The 3D printed vacuum manifold compatible with the LEGENDplex assay has been deposited in the NIH 3D print exchange (3DPX-021388).

All data needed to evaluate the conclusions in the paper are present in the paper, the Supplementary Materials, and the Figshare repository <http://doi.org/10.6084/m9.figshare.27091972>.

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Additional information

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI's ChatGPT to enhance clarity and flow of the writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Additional files

Supplemental Figures S1-S5 [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors have developed self-amplifying RNAs (saRNAs) encoding additional genes to suppress dsRNA-related inflammatory responses and cytokine release. Their results

demonstrate that saRNA constructs encoding anti-inflammatory genes effectively reduce cytotoxicity and cytokine production, enhancing the potential of saRNAs. This work is significant for advancing saRNA therapeutics by mitigating unintended immune activation.

Strengths:

This study successfully demonstrates the concept of enhancing saRNA applications by encoding immune-suppressive genes. A key challenge for saRNA-based therapeutics, particularly for non-vaccine applications, is the innate immune response triggered by dsRNA recognition. By leveraging viral protein properties to suppress immunity, the authors provide a novel strategy to overcome this limitation. The study presents a well-designed approach with potential implications for improving saRNA stability and minimizing inflammatory side effects.

Weaknesses:

(1) Impact on Cellular Translation:

The authors demonstrate that modified saRNAs with additional components enhance transgene expression by inhibiting dsRNA-sensing pathways. However, it is unclear whether these modifications influence global cellular translation beyond the expression of GFP and mScarlet-3 (which are encoded by the saRNA itself). Conducting a polysome profiling analysis or a puromycin labeling assay would clarify whether the modified saRNAs alter overall translation efficiency. This additional data would strengthen the conclusions regarding the specificity of dsRNA-sensing inhibition.

(2) Stability and Replication Efficiency of Long saRNA Constructs:

The saRNA constructs used in this study exceed 16 kb, making them more fragile and challenging to handle. Assessing their mRNA integrity and quality would be crucial to ensure their robustness.

Furthermore, the replicative capacity of the designed saRNAs should be confirmed. Since Figure 4 shows lower inflammatory cytokine production when encoding srIkB α and srIkB α -Smad7-SOCS1, it is important to determine whether this effect is due to reduced immune activation or impaired replication. Providing data on replication efficiency and expression levels of the encoded anti-inflammatory proteins would help rule out the possibility that reduced cytokine production is a consequence of lower replication.

(3) Comparative Data with Native saRNA:

Including native saRNA controls in Figures 5-7 would allow for a clearer assessment of the impact of additional genes on cytokine production. This comparison would help distinguish the effect of the encoded suppressor proteins from other potential factors.

(4) In vivo Validation and Safety Considerations:

Have the authors considered evaluating the in vivo potential of these saRNA constructs? Conducting animal studies would provide stronger evidence for their therapeutic applicability. If in vivo experiments have not been performed, discussing potential challenges - such as saRNA persistence, biodistribution, and possible secondary effects-would be valuable.

(5) Immune Response to Viral Proteins:

Since the inhibitors of dsRNA-sensing proteins (E3, NSs, and L*) are viral proteins, they would be expected to induce an immune response. Analyzing these effects in vivo would add insight into the applicability of this approach.

(6) Streamlining the Discussion Section:

The discussion is quite lengthy. To improve readability, some content - such as the rationale for gene selection-could be moved to the Results section. Additionally, the descriptions of Figure 3 should be consolidated into a single section under a broader heading for improved coherence.

<https://doi.org/10.7554/eLife.105978.1.sa1>

Reviewer #2 (Public review):

Summary:

Lim et al. have developed a self-amplifying RNA (saRNA) design that incorporates immunomodulatory viral proteins, and show that the novel design results in enhanced protein expression in vitro in mouse primary fibroblast-like synoviocytes. They test constructs including saRNA with the vaccinia virus E3 protein and another with E3, Toscana virus NS protein and Theiler's virus L protein (E3 + NS + L), and another with srIkB α -Smad7-SOCS1. They have also tested whether ML336, an antiviral, enables control of transgene expression.

Strengths:

The experiments are generally well-designed and offer mechanistic insight into the RNA-sensing pathways that confer enhanced saRNA expression. The experiments are carried out over a long timescale, which shows the enhance effect of the saRNA E3 design compared to the control. Furthermore, the inhibitors are shown to maintain the cell number, and reduce basal activation factor- α levels.

Weaknesses:

One limitation of this manuscript is that the RNA is not well characterized; some of the constructs are quite long and the RNA integrity has not been analyzed. Furthermore, for constructs with multiple proteins, it's imperative to confirm the expression of each protein to confirm that any therapeutic effect is from the effector protein (e.g. E3, NS, L). The ML336 was only tested at one concentration; it is standard in the field to do a dose-response curve. These experiments were all done in vitro in mouse cells, thus limiting the conclusion we can make about mechanisms in a human system.

<https://doi.org/10.7554/eLife.105978.1.sa0>

Author response:

Reviewer #1 (Public review):

Summary:

The authors have developed self-amplifying RNAs (saRNAs) encoding additional genes to suppress dsRNA-related inflammatory responses and cytokine release. Their results demonstrate that saRNA constructs encoding anti-inflammatory genes effectively reduce cytotoxicity and cytokine production, enhancing the potential of saRNAs. This work is significant for advancing saRNA therapeutics by mitigating unintended immune activation.

Strengths:

This study successfully demonstrates the concept of enhancing saRNA applications by encoding immune-suppressive genes. A key challenge for saRNA-based therapeutics, particularly for non-vaccine applications, is the innate immune response triggered by dsRNA recognition. By leveraging viral protein properties to suppress immunity, the authors provide a novel strategy to overcome this limitation. The study presents a well-designed approach with potential implications for improving saRNA stability and minimizing inflammatory side effects.

We thank Reviewer #1 for their thorough review and for recognizing both the significance of our work and the potential of our strategy to expand saRNA applications beyond vaccines.

Weaknesses:

(1) Impact on Cellular Translation:

The authors demonstrate that modified saRNAs with additional components enhance transgene expression by inhibiting dsRNA-sensing pathways. However, it is unclear whether these modifications influence global cellular translation beyond the expression of GFP and mScarlet-3 (which are encoded by the saRNA itself). Conducting a polysome profiling analysis or a puromycin labeling assay would clarify whether the modified saRNAs alter overall translation efficiency. This additional data would strengthen the conclusions regarding the specificity of dsRNA-sensing inhibition.

We thank the reviewer for this helpful insight and suggestion. We aim to conduct a puromycin labelling assay to clarify the effect of the various saRNA constructs on translation efficiency.

(2) Stability and Replication Efficiency of Long saRNA Constructs:

The saRNA constructs used in this study exceed 16 kb, making them more fragile and challenging to handle. Assessing their mRNA integrity and quality would be crucial to ensure their robustness.

Furthermore, the replicative capacity of the designed saRNAs should be confirmed. Since Figure 4 shows lower inflammatory cytokine production when encoding srIkBa and srIkBa-Smad7-SOCS1, it is important to determine whether this effect is due to reduced immune activation or impaired replication. Providing data on replication efficiency and expression levels of the encoded anti-inflammatory proteins would help rule out the possibility that reduced cytokine production is a consequence of lower replication.

This is another very helpful comment. We will conduct an analysis of saRNA integrity and quality by denaturing gel electrophoresis. To examine replicative capacity of the saRNA constructs, we aim to conduct RT-qPCR experiments.

(3) Comparative Data with Native saRNA:

Including native saRNA controls in Figures 5-7 would allow for a clearer assessment of the impact of additional genes on cytokine production. This comparison would help distinguish the effect of the encoded suppressor proteins from other potential factors.

Thank you for your suggestion. We will implement this change in the next version of the manuscript.

(4) In vivo Validation and Safety Considerations:

Have the authors considered evaluating the in vivo potential of these saRNA constructs? Conducting animal studies would provide stronger evidence for their therapeutic applicability. If in vivo experiments have not been performed, discussing potential challenges - such as saRNA persistence, biodistribution, and possible secondary effects- would be valuable.

(5) Immune Response to Viral Proteins:

Since the inhibitors of dsRNA-sensing proteins (E3, NSs, and L) are viral proteins, they would be expected to induce an immune response. Analyzing these effects in vivo would add insight into the applicability of this approach.*

We recognize the importance of in vivo studies and immune cell responses and plan to incorporate in vivo imaging in future studies to investigate these interactions, as well as examining delivery of various cargoes via saRNA to determine potential therapeutic benefits in different animal models of inflammatory pain, but such studies are beyond the scope of this current investigation. As suggested by the reviewer, we will incorporate a section on potential challenges of in vivo saRNA work in the revised manuscript.

(6) Streamlining the Discussion Section:

The discussion is quite lengthy. To improve readability, some content - such as the rationale for gene selection-could be moved to the Results section. Additionally, the descriptions of Figure 3 should be consolidated into a single section under a broader heading for improved coherence.

Thank you for your suggestions, we will make these changes in the next revision.

Reviewer #2 (Public review):

Summary:

Lim et al. have developed a self-amplifying RNA (saRNA) design that incorporates immunomodulatory viral proteins, and show that the novel design results in enhanced protein expression in vitro in mouse primary fibroblast-like synoviocytes. They test constructs including saRNA with the vaccinia virus E3 protein and another with E3, Toscana virus NS protein and Theiler's virus L protein (E3 + NS + L), and another with srIkBa-Smad7-SOCS1. They have also tested whether ML336, an antiviral, enables control of transgene expression.

Strengths:

The experiments are generally well-designed and offer mechanistic insight into the RNA-sensing pathways that confer enhanced saRNA expression. The experiments are carried out over a long timescale, which shows the enhance effect of the saRNA E3 design compared to the control. Furthermore, the inhibitors are shown to maintain the cell number, and reduce basal activation factor- α levels.

We thank Reviewer #2 for their detailed assessment and recognition of the mechanistic insights provided by our study.

Weaknesses:

One limitation of this manuscript is that the RNA is not well characterized; some of the constructs are quite long and the RNA integrity has not been analyzed. Furthermore, for constructs with multiple proteins, it's imperative to confirm the expression of each

protein to confirm that any therapeutic effect is from the effector protein (e.g. E3, NS, L). The ML336 was only tested at one concentration; it is standard in the field to do a dose-response curve. These experiments were all done in vitro in mouse cells, thus limiting the conclusion we can make about mechanisms in a human system.

We agree that these are weaknesses of our work. We plan to address some of these weaknesses by performing a dose response curve for ML336, examining saRNA integrity through denaturing gel electrophoresis, and will also aim to provide additional evidence for effects of effector proteins through RT-qPCR. We are also looking into testing these constructs in patient-derived FLS.

<https://doi.org/10.7554/eLife.105978.1.sa3>